The Dependency Ordered Formal Exposition of PDEs

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Abstract

This document is a step-indexed, dependency-ordered presentation of the theory of partial differential equations, following [1, 2] . Every result appears only after all results it depends on. Entries are split into postulates (unproved statements used inside proofs) and axioms of the ordered field $\mathbb{R}$ (primitive structure). The document covers the full prerequisite tower (measure theory, L^p spaces, Hilbert and Banach space theory, convex analysis, integration by parts) and the PDE solution layer built on it: Sobolev spaces and weak derivatives, existence, uniqueness and regularity for the Dirichlet problem for the Laplacian via Lax–Milgram, the weak maximum principle and Hopf lemma, the Dirichlet principle, weak existence and energy estimates for the heat and wave equations, and Duhamel's principle. Each entry is annotated with the choice strength of its proof: ZF, ZF+DC, or ZFC.

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Root theorems (nothing depends on them):
SpectralThm     CompactAdjoint   Minkowski
GenHolder       Interpolation    LebesgueDiff
Egoroff         Bochner          Pettis
SpectrumCompact   Jensen         ArzelaAscoli
GreenI          GreenIII         Polar
MovingRegions   MollifierProps   LebesgueExistence
ClosedGraph     ImplicitFT       DuhamelPrinciple
HeatEnergyDecay WaveWeakExistence HeatWeakExistence
StrongMaxPrinciple EllipticBoundaryReg EllipticInteriorReg
PoissonExistence PoissonUniqueness DirichletPrinciple
RellichKondrachov
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Deepest chains (by depth):
L0 | Minkowski
L1 |   Holder
L2 |     Young
L3 |       expconvex
L4 |         SecondDerivCriterion
L5 |           Taylor
L6 |             FTC
L7 |               MVT

L0 | HeatWeakExistence
L1 |   WeakCompact
L2 |     Kakutani
L3 |       BanachAlaoglu
L4 |         Tych

L0 | RellichKondrachov
L1 |   WeakCompact
L2 |     Kakutani
L3 |       BanachAlaoglu
L4 |         Tych

L0 | SpectralThm
L1 |   BoundsSpectrum
L2 |     LaxMilgram
L3 |       Riesz
L4 |         {Sperp, perp}
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Figure 1. Summary of root theorems and the deepest dependency chains.

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Key words and phrases. notation, inequalities, Gauss–Green, mollifiers, Banach space, Hilbert space, weak convergence, Lebesgue measure, Sobolev spaces, weak derivatives, weak solutions, elliptic regularity, maximum principle, heat equation, wave equation, Morrey, trace, semigroup.

1. Introduction

The document is organised into six parts: Axioms, Postulates, Definitions, Statements (Lemmas, Theorems), Proofs, and Scripts. Following the reverse-mathematics convention of [5], entries are split into two primitive layers: axioms of the ordered field $\mathbb{R}$ and postulates (unproved statements admitted inside proofs). Every entry in a later part may depend only on earlier entries, and no entry may depend on anything outside the union of axioms and postulates. Lemmas and Theorems form a single dependency layer: an entry may depend on an earlier Lemma or Theorem, so long as the global order is respected.

Choice-strength tags record the strength needed to make the given proof go through; they are upper bounds on the logical strength of each entry and are not claimed to be tight.

2. Axioms

The axioms of the ordered field $\mathbb{R}$, kept as primitive structure.

Ax 1	ax : R − comm : $\forall(x:\mathbb{R}),\ \forall(y:\mathbb{R}),\ x+y=y+x$	AdditiveCommutativity	ZF
Ax 2	ax : R − assoc : $\forall(x:\mathbb{R}),\ \forall(y:\mathbb{R}),\ \forall(z:\mathbb{R}),\ (x+y)+z=x+(y+z)$	AdditiveAssociativity	ZF
Ax 3	ax : R − order : $\forall(x:\mathbb{R}),\ \forall(y:\mathbb{R}),\ \forall(z:\mathbb{R}),\ x\le y\wedge y\le z\Rightarrow x\le z$	OrderTransitivity	ZF
Ax 4	ax : R − complete : $\forall(a:\mathbb{N}\to\mathbb{R}),\ \mathsf{Cauchy}(a)\Rightarrow\exists(L:\mathbb{R}),\ \lim_n a(n)=L$	CompletenessofR	ZF
Ax 5	ax : R − mul : $\forall(x:\mathbb{R}),\ \forall(y:\mathbb{R}),\ \forall(z:\mathbb{R}),\ x\le y\wedge 0\le z\Rightarrow xz\le yz$	MultiplicationCompat.	ZF
Ax 6	ax : R − add : $\forall(x:\mathbb{R}),\ \forall(y:\mathbb{R}),\ \forall(z:\mathbb{R}),\ x\le y\Rightarrow x+z\le y+z$	AdditionCompat.	ZF

3. Postulates

Unproved statements admitted inside proofs (postulates).

Pos 1	MVT : $\forall(f:\mathbb{R}\to\mathbb{R}),\ C([a,b])(f)\wedge D((a,b))(f)\Rightarrow\exists(c:(a,b)),\ f'(c)=(f(b)-f(a))/(b-a)$	MeanValueTheorem	ZF
Pos 2	MVTI : $\forall(w:\mathbb{R}\to\mathbb{R}),\ \forall(g:\mathbb{R}\to\mathbb{R}),\ w\ge 0\wedge C([a,b])(g)\Rightarrow\exists(c:[a,b]),\ \int_a^b w(t)g(t)\,dt=g(c)\int_a^b w(t)\,dt$	WeightedMeanValue	ZF
Pos 3	IBP : $\forall(u:\mathbb{R}\to\mathbb{R}),\ \forall(v:\mathbb{R}\to\mathbb{R}),\ C^1(u)\wedge C^1(v)\Rightarrow\int_a^b u'(t)v(t)\,dt=[uv]_a^b-\int_a^b u(t)v'(t)\,dt$	IntegrationbyParts	ZF
Pos 4	BW : $\forall(x:\mathbb{N}\to\mathbb{R}^n),\ \mathsf{Bdd}(x)\Rightarrow\exists(x':\mathbb{N}\to\mathbb{R}^n),\ \exists(L:\mathbb{R}^n),\ \mathsf{Subseq}(x',x)\wedge x'\to L$	Bolzano − Weierstrass	ZF
Pos 5	Tych : $\forall(X:I\to\mathsf{Top}),\ (\forall(i:I),\ \mathsf{Compact}(X_i))\Rightarrow\mathsf{Compact}(\prod_i X_i)$	Tychonoff	ZFC
Pos 6	Morrey : $\forall(p>n),\ \forall(u\in W^{1,p}(U)),\ \exists(C_M:\mathbb{R}_{>0}),\ \|u\|_{C^{0,1-n/p}(U)}\le C_M\|u\|_{W^{1,p}(U)}$	Morrey	ZF + DC

Pos 7	**Trace :** $\exists(\mathsf{Tr} : H^1(U) \to L^2(\partial U)),\ \exists(\mathsf{Ext} : H^{3/2}(\partial U) \to H^2(U)),\ \mathsf{lin.\ cont.}(\mathsf{Tr}) \wedge \mathsf{lin.\ cont.}(\mathsf{Ext}) \wedge \mathsf{Tr} \circ \mathsf{Ext} = \mathsf{id}$	**BoundedTrace/Ext**	**ZF + DC**
Pos 8	**EllipticBoundaryReg-Step :** $\forall(\partial U \in C^2,\ f \in L^2,\ g \in H^{3/2}(\partial U)),\ \exists!(u \in H^2(U)),\ -\Delta u = f \wedge \mathsf{Tr}(u) = g \wedge \|u\|_{H^2} \leq C(\|f\|_{L^2} + \|g\|_{H^{3/2}})$	**BoundaryRegularity Step**	**ZF + DC**
Pos 9	**HeatSemigroupGenerator :** $\exists(S : [0,\infty) \to \mathcal{L}(L^2)),\ S\ \mathsf{C_0\text{-}semigroup},\ S'(t) = \Delta S(t)\ \mathsf{on}\ H_0^1 \cap H^2$	**HeatSemigroup Generator**	**ZF + DC**

4. Definitions

Def 1	**Def : M :** $\mathrm{M}^{m\times n} := \mathbb{R}^{m\times n}$	**MatrixSpace**	**ZF**
Def 2	**Def : S :** $\mathrm{S}^n := \{A : \mathrm{M}^{n\times n} \mid A^T = A\}$	**SymmetricMatrixSpace**	**ZF**
Def 3	**Def : tr :** $\forall(A : \mathrm{M}^{n\times n}),\ \mathsf{tr}(A) = \sum_{i=1}^n A_{ii}$	**Trace**	**ZF**
Def 4	**Def : diag :** $\forall(d : \mathbb{N} \to \mathbb{R}),\ \forall(i : \mathbb{N}),\ \forall(j : \mathbb{N}),\ \mathsf{diag}(d)_{ij} = d_i\delta_{ij}$	**DiagonalMatrix**	**ZF**
Def 5	**Def : colon :** $\forall(A : \mathrm{M}^{m\times n}),\ \forall(B : \mathrm{M}^{m\times n}),\ \mathsf{colon}(A,B) = \sum_{i,j} A_{ij}B_{ij}$	**MatrixInnerProduct**	**ZF**
Def 6	**Def : quad :** $\forall(A : \mathrm{S}^n),\ \forall(x : \mathbb{R}^n),\ \mathsf{quad}(A,x) = \sum_{i,j} A_{ij}x_ix_j$	**QuadraticForm**	**ZF**
Def 7	**Def : sphere :** $S^{n-1} := \partial B(0,1) \subset \mathbb{R}^n$	**UnitSphere**	**ZF**
Def 8	**Def : halfspace :** $\mathbb{R}^n_+ := \{x : \mathbb{R}^n \mid x_n > 0\}$	**UpperHalf − Space**	**ZF**
Def 9	**Def : ccontained :** $\forall(U : \mathsf{Set}),\ \forall(V : \mathsf{Set}),\ \mathsf{ccontained}(V,U) \Leftrightarrow V \subset \bar{V} \subset U \wedge \mathsf{compact}(\bar{V})$	**CompactContainment**	**ZF**
Def 10	**Def : UT :** $U_T := U \times (0,T]$	**ParabolicCylinder**	**ZF**
Def 11	**Def : alpha :** $\alpha(n) := |B(0,1)| = \pi^{n/2}/\Gamma(n/2+1)$	**UnitBallVolume**	**ZF**
Def 12	**Def : spt :** $\forall(u : \mathbb{R}^n \to \mathbb{R}),\ \mathsf{spt}(u) := \overline{\{x : \mathbb{R}^n \mid u(x) \neq 0\}}$	**Support**	**ZF**
Def 13	**Def : posneg :** $\forall(u : \mathbb{R} \to \mathbb{R}),\ \mathsf{pos}(u) = \max(u,0) \wedge \mathsf{neg}(u) = -\min(u,0)$	**Positive/NegativeParts**	**ZF**
Def 14	**Def : sgn :** $\forall(x : \mathbb{R}),\ \mathsf{sgn}(x) = 1_{\{x>0\}} - 1_{\{x<0\}}$	**SignFunction**	**ZF**
Def 15	**Def : char :** $\forall(E : \mathsf{Set}),\ \forall(x : \mathbb{R}),\ \mathsf{char}(E,x) = 1_{\{x\in E\}}$	**CharacteristicFunction**	**ZF**
Def 16	**Def : Lip :** $\forall(u : \mathbb{R}^n \to \mathbb{R}),\ \mathsf{Lip}(u) = \sup_{x\neq y} |u(x)-u(y)|/|x-y|$	**LipschitzConstant**	**ZF**
Def 17	**Def : conv :** $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(g : \mathbb{R}^n \to \mathbb{R}),\ f,g\ \mathsf{meas.},\ (f*g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)\,dy$	**Convolution**	**ZF**
Def 18	**Def : multi :** $\forall(\alpha : \mathbb{N}^n),\ |\alpha| = \alpha_1 + \cdots + \alpha_n$	**Multi − IndexOrder**	**ZF**
Def 19	**Def : Dalpha :** $\forall(\alpha : \mathbb{N}^n),\ D^\alpha u = \partial_{x_1}^{\alpha_1} \dots \partial_{x_n}^{\alpha_n} u$	**Multi − Index Derivative**	**ZF**
Def 20	**Def : grad :** $\nabla u = (u_{x_1}, \dots, u_{x_n})$	**Gradient**	**ZF**

$\mathbf{Def\,21}$	Def : bigO : $\forall(f : \mathbb{R} \to \mathbb{R}),\ \forall(g : \mathbb{R} \to \mathbb{R}),\ f = O(g) \Leftrightarrow \exists(C : \mathbb{R}),\ \forall(x : \mathbb{R}),\ \lvert f(x)\rvert \leq C\lvert g(x)\rvert$	Big − ONotation	ZF
$\mathbf{Def\,22}$	Def : littleo : $\forall(f : \mathbb{R} \to \mathbb{R}),\ \forall(g : \mathbb{R} \to \mathbb{R}),\ f = o(g) \Leftrightarrow \lim_{x\to 0} \lvert f(x)\rvert/\lvert g(x)\rvert = 0$	Little − oNotation	ZF
$\mathbf{Def\,23}$	Def : moll : $\forall(x : \mathbb{R}^n),\ \eta(x) = C\exp(1/(\lvert x\rvert^2 - 1))1_{\lvert x\rvert<1} \wedge \int \eta = 1$	Mollifier	ZF
$\mathbf{Def\,24}$	Def : Norm : $\lVert\cdot\rVert$ satisfies $\triangle$, homog, sep	Norm	ZF
$\mathbf{Def\,25}$	Def : nu : $\nu = (\nu^1, \ldots, \nu^n) \wedge \lvert\nu\rvert = 1$	UnitNormal	ZF
$\mathbf{Def\,26}$	Def : funcCk : $\forall(k : \mathbb{N}),\ \forall(U : \mathrm{Set}),\ C^k(U) = \{u : U \to \mathbb{R} \mid k\text{-times cont. diff.}\}$	CkFunctions	ZF
$\mathbf{Def\,27}$	Def : normM : $\forall(A : \mathrm{M}^{m\times n}),\ \lVert A\rVert = (\mathrm{colon}(A, A))^{1/2}$	MatrixNorm	ZF
$\mathbf{Def\,28}$	Def : ge : $\forall(A : \mathrm{S}^n),\ \forall(B : \mathrm{S}^n),\ A \geq B \Leftrightarrow A - B$ nonneg.	MatrixOrder	ZF
$\mathbf{Def\,29}$	Def : parab : $\Gamma_T := \bar{U}_T \setminus U_T$	ParabolicBoundary	ZF
$\mathbf{Def\,30}$	Def : avg − ball : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(x : \mathbb{R}^n),\ \forall(r : \mathbb{R}_{>0}),\ \mathrm{avgB}(f, x, r) := \frac{1}{\alpha(n)r^n} \int_{B(x,r)} f\,dy$	BallAverage	ZF
$\mathbf{Def\,31}$	Def : avg − sph : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(x : \mathbb{R}^n),\ \forall(r : \mathbb{R}_{>0}),\ \mathrm{avgS}(f, x, r) = \frac{1}{n\alpha(n)r^{n-1}} \int_{\partial B(x,r)} f\,dS$	SphereAverage	ZF
$\mathbf{Def\,32}$	Def : avg − set : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(E : \mathrm{Set}),\ \mathrm{avgE}(f, E) = \frac{1}{\mu(E)} \int_E f\,d\mu \wedge \mu(E) > 0$	SetAverage	ZF + DC
$\mathbf{Def\,33}$	Def : Simple : $\forall(N : \mathbb{N}),\ \exists(E : \{1, \ldots, N\} \to \mathcal{M}),\ \exists(u : \{1, \ldots, N\} \to \mathbb{R}),\ \mathrm{s} = \sum_{i=1}^N \chi_{E_i} u_i$	SimpleFunction	ZF
$\mathbf{Def\,34}$	Def : StrongConv : $\forall(u : \mathbb{N} \to H),\ \forall(u_\infty : H),\ u_k \to u_\infty \Leftrightarrow \lim \lVert u_k - u_\infty\rVert = 0$	StrongConvergence	ZF
$\mathbf{Def\,35}$	Def : perp : $\forall(u : H),\ \forall(v : H),\ (u, v) = 0$	Orthogonality	ZF
$\mathbf{Def\,36}$	Def : ONB : $\forall(w : \mathbb{N} \to H),\ \mathrm{ONB}(w) \Leftrightarrow (\forall k \neq l,\ (w_k, w_l) = 0) \wedge (\forall k,\ \lVert w_k\rVert = 1) \wedge (\forall u,\ (\forall k,\ (u, w_k) = 0) \Rightarrow u = 0)$	OrthonormalBasis	ZF
$\mathbf{Def\,37}$	Def : Sperp : $S^\perp = \{u : H \mid \forall(v : H),\ (u, v) = 0\}$	Orthogonal Complement	ZF
$\mathbf{Def\,38}$	Def : Linear : $\forall(A : H \to H),\ \forall(u : H),\ \forall(v : H),\ \forall(\lambda : \mathbb{R}),\ \forall(\mu : \mathbb{R}),\ A(\lambda u + \mu v) = \lambda Au + \mu Av$	LinearOperator	ZF
$\mathbf{Def\,39}$	Def : Bdd : $\forall(A : H \to H),\ \exists(C : \mathbb{R}),\ \forall(u : H),\ \lVert Au\rVert \leq C\lVert u\rVert$	BoundedOperator	ZF
$\mathbf{Def\,40}$	Def : ClosedOp : $\forall(A : H \to H),\ \forall(u : \mathbb{N} \to H),\ \forall(u_\infty : H),\ \forall(v : H),\ u_k \to u_\infty \wedge Au_k \to v \Rightarrow Au_\infty = v$	ClosedOperator	ZF
$\mathbf{Def\,41}$	Def : rho : $\forall(A : H \to H),\ \rho(A) = \{\eta : \mathbb{R} \mid \exists(B : H \to H),\ \mathrm{bdd}(B) \wedge B = (A - \eta I)^{-1}\}$	ResolventSet	ZF
$\mathbf{Def\,42}$	Def : pairing : $\forall(u^* : X^*),\ \forall(u : X),\ u^*(u)$	DualPairing	ZF
$\mathbf{Def\,43}$	Def : Hess : $D^2u = (u_{x_i x_j})_{ij}$	Hessian	ZF
$\mathbf{Def\,44}$	Def : Dk : $\forall(k : \mathbb{N}),\ D^k u = \{D^\alpha u : \mathbb{N}^n \mid \lvert\alpha\rvert = k\}$	k − thDerivativeSet	ZF
$\mathbf{Def\,45}$	Def : dnu : $\partial u/\partial\nu = \nu \cdot \nabla u$	NormalDerivative	ZF
$\mathbf{Def\,46}$	Def : moll − eps : $\forall(\epsilon : \mathbb{R}_{>0}),\ \forall(x : \mathbb{R}^n),\ \eta_\epsilon(x) = \epsilon^{-n}\eta(x/\epsilon)$	StandardMollifier	ZF

$\mathbf{Def\,47}$	Def : InnerProduct : $(\cdot,\cdot)$ sym., bil., pos.	InnerProduct	ZF
$\mathbf{Def\,48}$	Def : Lap : $\Delta u = \sum_i u_{x_i x_i} = \mathrm{tr}(D^2 u)$	Laplacian	ZF
$\mathbf{Def\,49}$	Def : Ckbnd : ∂U loc. graph of C^k fn.	CkBoundary	ZF
$\mathbf{Def\,50}$	Def : LebesguePoint : $\forall(f:\mathbb{R}^n\to\mathbb{R}),\ \forall(x_0:\mathbb{R}^n),\ \lim_{r\to 0}\mathrm{avgB}(\lvert f-f(x_0)\rvert,x_0,r)=0$	LebesguePoint	ZF + DC
$\mathbf{Def\,51}$	Def : moll $-$ f : $\forall(\epsilon:\mathbb{R}_{>0}),\ f^\epsilon=\eta_\epsilon * f$	Mollification	ZF + DC
$\mathbf{Def\,52}$	Def : Dknorm : $\forall(k:\mathbb{N}),\ \lvert D^k u\rvert=(\sum_{\lvert\alpha\rvert=k}\lvert D^\alpha u\rvert^2)^{1/2}$	k $-$ thDerivativeNorm	ZF
$\mathbf{Def\,53}$	Def : StronglyMeasurable : $\forall(\mathrm{f}:X),\ (\exists(\mathrm{s}:\mathbb{N}\to X),\ \mathrm{simple}(\mathrm{s}_k)\wedge \mathrm{s}_k\to \mathrm{f}$ a.e.)	StrongMeasurability	ZF + DC
$\mathbf{Def\,54}$	Def : Cauchy : $\forall(u:\mathbb{N}\to H),\ \forall(\epsilon:\mathbb{R}_{>0}),\ \exists(N:\mathbb{N}),\ \forall(k:\mathbb{N}),\ \forall(l:\mathbb{N}),\ k,l\geq N\Rightarrow \lVert u_k-u_l\rVert<\epsilon$	CauchySequence	ZF
$\mathbf{Def\,55}$	Def : Eigenvalue : $\forall(A:H\to H),\ \forall(\eta:\mathbb{R}),\ N(A-\eta I)\neq\{0\}$	Eigenvalue	ZF
$\mathbf{Def\,56}$	Def : sigma : $\forall(A:H\to H),\ \sigma(A)=\mathbb{R}\setminus\rho(A)$	Spectrum	ZF
$\mathbf{Def\,57}$	Def : Xstar : $X^*=\{\phi:H\to\mathbb{R}\mid \text{lin.}\wedge\text{bdd.}\}$	DualSpace	ZF
$\mathbf{Def\,58}$	Def : SigmaAlgebra : $\mathcal{M}$ closed under complement and countable unions	Sigma $-$ Algebra	ZF
$\mathbf{Def\,59}$	Def : Measurable : $\forall(f:X\to\mathbb{R}),\ \forall(U:\text{Open}),\ f^{-1}(U)\in\mathcal{M}$	MeasurableFunction	ZF
$\mathbf{Def\,60}$	Def : Summable : $\forall(f:X\to\mathbb{R}),\ \int\lvert f\rvert<\infty$	SummableFunction	ZF + DC
$\mathbf{Def\,61}$	Def : esssup : $\forall(f:X\to\mathbb{R}),\ \inf\{\mu:\mathbb{R}\mid\lvert\{f>\mu\}\rvert=0\}$	EssentialSupremum	ZF + DC
$\mathbf{Def\,62}$	Def : Banach : complete normed lin. space	BanachSpace	ZF
$\mathbf{Def\,63}$	Def : Hilbert : $\forall(H:\text{Banach}),\ \exists(\cdot,\cdot),\ \lVert u\rVert^2=(u,u)$	HilbertSpace	ZF
$\mathbf{Def\,64}$	Def : Convex : $\forall(f:\mathbb{R}^n\to\mathbb{R}),\ \forall(x:\mathbb{R}^n),\ \forall(y:\mathbb{R}^n),\ \forall(\tau:[0,1]),\ f(\tau x+(1-\tau)y)\leq\tau f(x)+(1-\tau)f(y)$	ConvexFunction	ZF
$\mathbf{Def\,65}$	Def : Reflexive : $(X^*)^*=X$	ReflexiveSpace	ZF
$\mathbf{Def\,66}$	Def : WeakConv : $\forall(u:\mathbb{N}\to X),\ \forall(u_\infty:X),\ u_k\rightharpoonup u_\infty\Leftrightarrow\forall(u^*:X^*),\ u^*(u_k)\to u^*(u_\infty)$	WeakConvergence	ZF
$\mathbf{Def\,67}$	Def : adjoint : $\forall(A:H\to H),\ \forall(u:H),\ \forall(v:H),\ (Au,v)=(u,A^*v)$	AdjointOperator	ZF
$\mathbf{Def\,68}$	Def : WeaklyMeasurable : $\forall(\mathrm{f}:X),\ (\forall(u^*:X^*),\ (t\mapsto u^*(\mathrm{f}(t)))$ meas.)	WeakMeasurability	ZF + DC
$\mathbf{Def\,69}$	Def : Separable : $\exists(D:\text{Set}),\ \text{countable}(D)\wedge\text{dense}(D)$	SeparableSpace	ZF
$\mathbf{Def\,70}$	Def : AlmostSeparablyValued : $\forall(\mathrm{f}:X),$ separable$(\lvert\{\mathrm{f}(t):t\in\mathbb{R}\}\rvert)$ off null set	AlmostSep.Valued	ZF + DC
$\mathbf{Def\,71}$	Def : Symmetric : $\forall(A:H\to H),\ A^*=A$	SymmetricOperator	ZF
$\mathbf{Def\,72}$	Def : CompactOp : $\forall(A:H\to H),$ bdd seq $\to$ precompact image	CompactOperator	ZF
$\mathbf{Def\,73}$	Def : CompCont : $\forall(C:\text{Set}),\ \forall(U:\text{Set}),\ C\subset\subset U\Leftrightarrow\bar{C}$ compact $\wedge\ \bar{C}\subset U$	CompactContained	ZF
$\mathbf{Def\,74}$	Def : Distribution : $\mathcal{D}(U):=\{\phi\in C^\infty(U):$ compact supp$(\phi)\}$	TestFunctionSpace	ZF

Def 75	**Def : WeakDeriv** : $\forall(u \in L^1_{\text{loc}}(U)),\ \forall(\alpha : \mathbb{N}^n),\ \forall(v \in L^1_{\text{loc}}(U)),\ \text{WeakDeriv}^\alpha(u) = v \Leftrightarrow \forall(\phi \in \mathcal{D}(U)),\ \int_U v\phi = (-1)^{\lvert\alpha\rvert} \int_U uD^\alpha\phi$	**WeakDerivative**	**ZF + DC**
Def 76	**Def : SobolevWkp** : $\forall(k : \mathbb{N}),\ \forall(p \in [1,\infty]),\ W^{k,p}(U) := \{u \in L^p(U) : \forall(\alpha : \mathbb{N}^n),\ \lvert\alpha\rvert \le k \Rightarrow \exists(v \in L^p(U)),\ \text{WeakDeriv}^\alpha(u) = v\}$	**SobolevSpace**	**ZF + DC**
Def 77	**Def : SobolevHk** : $\forall(k : \mathbb{N}),\ H^k(U) := W^{k,2}(U)$	**SobolevHilbertSpace**	**ZF + DC**
Def 78	**Def : H01** : $H^1_0(U) := \overline{\mathcal{D}(U)}^{\lVert\cdot\rVert_{H^1}}$	**ZeroTraceSobolev**	**ZF + DC**
Def 79	**Def : EllipticForm** : $B[u,v] := \int_U \nabla u \cdot \nabla v$	**DirichletForm**	**ZF + DC**
Def 80	**Def : Elliptic** : $\forall(a^{ij} \in L^\infty(U)),\ \text{Elliptic}(a) \Leftrightarrow \exists(\theta > 0),\ \forall(\xi \in \mathbb{R}^n),\ a^{ij}\xi_i\xi_j \ge \theta\lvert\xi\rvert^2$	**UniformEllipticity**	**ZF**
Def 81	**Def : PoissonWeakSol** : $\forall(f \in H^{-1}(U)),\ u \in H^1_0(U)$ solves $-\Delta u = f \Leftrightarrow \forall(v \in H^1_0(U)),\ B[u,v] = \langle f, v\rangle$	**WeakSolutionDirichlet**	**ZF + DC**
Def 82	**Def : ParabolicWeakSol** : $u \in L^2(0,T;H^1_0(U)) \wedge u_t \in L^2(0,T;H^{-1}(U))$ solves $u_t - \Delta u = f \Leftrightarrow \forall(v \in H^1_0),\ \langle u_t, v\rangle + B[u,v] = \langle f, v\rangle$ a.e. t	**WeakSolutionHeat**	**ZF + DC**
Def 83	**Def : WaveWeakSol** : $u \in L^2(0,T;H^1_0(U)) \wedge u_t \in L^2(0,T;L^2(U)) \wedge u_{tt} \in L^2(0,T;H^{-1}(U))$ solves $u_{tt} - \Delta u = f \Leftrightarrow \forall(v \in H^1_0),\ \langle u_{tt}, v\rangle + B[u,v] = \langle f, v\rangle$ a.e. t	**WeakSolutionWave**	**ZF + DC**
Def 84	**Def : DirichletEnergy** : $I[w] := \frac{1}{2}\int_U \lvert\nabla w\rvert^2 - \int_U fw,\ w \in H^1_0(U)$	**DirichletEnergy**	**ZF + DC**
Def 85	**Def : Harmonic** : $\forall(u \in C^2(U)),\ \text{Harmonic}(u) \Leftrightarrow \Delta u = 0$ in U	**HarmonicFunction**	**ZF**
Def 86	**Def : FundamentalSolution** : $\Delta\Phi = \delta_0\Phi(x) := \begin{cases} -\frac{1}{2\pi}\log\lvert x\rvert, & \text{if } n = 2 \\ \frac{1}{n(n-2)\alpha(n)}\frac{1}{\lvert x\rvert^{n-2}}, & \text{if } n \ge 3 \end{cases}$	**FundamentalSolution** ,	**ZF**
Def 87	**Def : Conormal** : $\partial u/\partial\nu_a := a^{ij}u_{x_i}\nu^j$	**Co − normalDerivative**	**ZF**

5. Statements

5.1 Lemmas

Lem 1	**FTC** : $\forall(f : \mathbb{R} \to \mathbb{R}),\ C^1([a,b])(f) \Rightarrow \int_a^b f' = f(b) - f(a)$	**FundamentalThmof Calculus**	**ZF**
Lem 2	**Taylor** : $\forall(f : \mathbb{R} \to \mathbb{R}),\ C^2(f) \Rightarrow \exists(\xi : [\min(x,z), \max(x,z)]),\ f(x) = f(z) + f'(z)(x-z) + \frac{1}{2}f''(\xi)(x-z)^2$	**TaylorwithLagrange Remainder**	**ZF**
Lem 3	**ProductRule** : $\forall(u : \mathbb{R} \to \mathbb{R}),\ \forall(v : \mathbb{R} \to \mathbb{R}),\ (uv)' = u'v + uv'$	**ProductRule**	**ZF**
Lem 4	**ChainRule** : $\forall(f : \mathbb{R} \to \mathbb{R}),\ \forall(g : \mathbb{R} \to \mathbb{R}),\ (f \circ g)' = f'(g)g'$	**ChainRule**	**ZF**
Lem 5	**JacobiFormula** : $\forall(A : \mathbb{R} \to \mathrm{M}^{n\times n}),\ C^1(A) \Rightarrow \forall(t : \mathbb{R}),\ \det A(t) \ne 0 \Rightarrow \frac{d}{dt}\det A = \det A \cdot \mathrm{tr}(A^{-1}A')$	**JacobiFormula (invertible)**	**ZF**
Lem 6	**JacobiFormulaAdj** : $\forall(A : \mathbb{R} \to \mathrm{M}^{n\times n}),\ C^1(A) \Rightarrow \frac{d}{dt}\det A = \mathrm{tr}(\mathrm{adj}(A)A')$	**JacobiFormula (adjugate)**	**ZF**
Lem 7	**Cauchy** : $\forall(a : \mathbb{R}),\ \forall(b : \mathbb{R}),\ ab \le \frac{1}{2}a^2 + \frac{1}{2}b^2$	**CauchyInequality**	**ZF**
Lem 8	**CauchyEps** : $\forall(a : \mathbb{R}),\ \forall(b : \mathbb{R}),\ \forall(\epsilon : \mathbb{R}_{>0}),\ ab \le \epsilon a^2 + \frac{b^2}{4\epsilon}$	**CauchyIneq.withϵ**	**ZF**

Lem 9	SecondDerivCriterion : $\forall(f : \mathbb{R} \to \mathbb{R}),\ C^2(f) \wedge f'' \geq 0 \Rightarrow f$ convex	SecondDeriv.Convexity	ZF
Lem 10	expconvex : $\forall(x : \mathbb{R}),\ \forall(y : \mathbb{R}),\ \forall(\tau : [0,1]),\ e^{\tau x+(1-\tau)y} \leq \tau e^x + (1-\tau)e^y$	ExponentialConvexity	ZF
Lem 11	Young : $\forall(p : \mathbb{R}_{>1}),\ \forall(q : \mathbb{R}_{>1}),\ 1/p + 1/q = 1 \Rightarrow \forall(a : \mathbb{R}_{>0}),\ \forall(b : \mathbb{R}_{>0}),\ ab \leq a^p/p + b^q/q$	YoungInequality	ZF
Lem 12	YoungEps : $\forall(p : \mathbb{R}_{>1}),\ \forall(q : \mathbb{R}_{>1}),\ 1/p + 1/q = 1 \Rightarrow \forall(a : \mathbb{R}_{>0}),\ \forall(b : \mathbb{R}_{>0}),\ \forall(\epsilon : \mathbb{R}_{>0}),\ ab \leq \epsilon a^p + C(\epsilon)b^q$	Youngwithϵ	ZF
Lem 13	CauchySchwarz : $\forall(x : \mathbb{R}^n),\ \forall(y : \mathbb{R}^n),\ \lvert x \cdot y\rvert \leq \lvert x\rvert\lvert y\rvert$	Cauchy − Schwarz	ZF
Lem 14	CauchySchwarzA : $\forall(A : \mathbf{S}^n),\ A > 0 \wedge A^T = A \Rightarrow \forall(x : \mathbb{R}^n),\ \forall(y : \mathbb{R}^n),\ \lvert x^T A y\rvert \leq (x^T A x)^{1/2}(y^T A y)^{1/2}$	Cauchy − Schwarzfor Matrices	ZF
Lem 15	BanachFP : $\forall(T : X \to X),$ contraction$(T) \wedge$ complete$(X) \Rightarrow \exists!(x^* : X),\ Tx^* = x^*$	BanachFixedPoint	ZF
Lem 16	SepHyper : $\forall(C :$ Set$),$ convex$(C) \wedge$ closed$(C) \Rightarrow \forall(p : \partial C),\ \exists(u : \mathbb{R}^n),\ u \neq 0 \wedge \forall(z : C),\ u \cdot (z - p) \leq 0$	SeparatingHyperplane	ZF
Lem 17	Baire : complete metric $\Rightarrow$ countable intersection of dense open is dense	BaireCategory	ZF + DC
Lem 18	OpenMap : $\forall(T : X \to Y),$ surjective$(T) \wedge$ bdd$(T) \Rightarrow$ open(T)	OpenMapping	ZF + DC
Lem 19	HardyLittlewood : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ L^1_{\text{loc}}(f) \Rightarrow$ avgB$(\lvert f - f(x)\rvert, x, r) \to 0$ a.e.	Hardy − Littlewood	ZF + DC
Lem 20	Riesz : $\forall(H :$ Hilbert$),\ \forall(u^* : H^*),\ \exists!(u : H),\ \forall(v : H),\ u^*(v) = (u, v)$	RieszRepresentation	ZF + DC
Lem 21	LaxMilgram : $\forall(a : H \to H \to \mathbb{R}),$ coercive$(a) \wedge$ bounded$(a) \Rightarrow \forall(f : H^*),\ \exists!(u : H),\ \forall(v : H),\ a(u, v) = f(v)$	Lax − Milgram	ZF + DC
Lem 22	BoundsSpectrum : $\forall(S : H \to H),\ \forall(m : \mathbb{R}),\ \forall(M : \mathbb{R}),\ \sigma(S) \subset [m, M] \wedge m \in \sigma(S) \wedge M \in \sigma(S)$	BoundsonSpectrum	ZF + DC
Lem 23	WeakDerivRules : $\forall(u, w \in W^{1,p}(U)),\ \forall(\phi \in \mathcal{D}(U)),$ WeakDeriv$(D(u + w)) = Du + Dw \wedge$ WeakDeriv$(D(u\phi)) = u'\phi + uD\phi$	WeakDerivativeRules	ZF + DC
Lem 24	Poincare : $\forall(\partial U \in C^0),\ \forall(u \in H^1_0(U)),\ \lVert u\rVert_{L^2(U)} \leq C_P(U)\lVert \nabla u\rVert_{L^2(U)}$	PoincaréInequality	ZF + DC
Lem 25	TraceInequality : $\forall(u \in H^1(U)),\ \lVert u\rVert_{L^2(\partial U)} \leq C_T\lVert u\rVert_{H^1(U)}$	TraceInequality	ZF + DC
Lem 26	SobolevEmbedding : $\forall(u \in W^{1,p}(U)),\ p > n \Rightarrow u \in C^{0,1-n/p}(U) \wedge \lVert u\rVert_{C^{0,1-n/p}} \leq C\lVert u\rVert_{W^{1,p}}$	SobolevEmbedding	ZF + DC
Lem 27	EnergyEstimateElliptic : $\forall(f \in L^2(U)),\ u$ weak sol $\Rightarrow \lVert u\rVert_{H^1_0} \leq C\lVert f\rVert_{L^2}$	EllipticEnergyEstimate	ZF + DC
Lem 28	EllipticInteriorReg-Step : $\forall(u \in H^1,\ -\Delta u = f \in L^2),\ \forall(V \subset\subset U),\ \lVert u\rVert_{H^2(V)} \leq C(\lVert f\rVert_{L^2} + \lVert u\rVert_{H^1})$	InteriorRegularityStep	ZF + DC

5.2 Theorems

Thm 1	BanachAlaoglu : $\forall(X :$ Banach$),\ \overline{B}_{X^*}(0, 1)$ weak*-compact	Banach − Alaoglu	ZFC
Thm 2	Kakutani : $\forall(X :$ Banach$),$ reflexive$(X) \Rightarrow \overline{B}_X(0, 1)$ weakly compact	Kakutani	ZFC
Thm 3	InverseFT : $\forall(f : \mathbb{R}^n \to \mathbb{R}^n),\ Jf(x_0) \neq 0 \Rightarrow f$ locally invertible $\wedge\ f^{-1} \in C^1$	InverseFunction	ZF
Thm 4	ImplicitFT : $\forall(f : \mathbb{R}^n \times \mathbb{R}^m \to \mathbb{R}^m),\ J_y f(x_0, y_0) \neq 0 \Rightarrow \exists(g : \mathbb{R}^n \to \mathbb{R}^m),\ f(x, g(x)) = z_0$	ImplicitFunction	ZF

Thm 5	ClosedGraph : $\forall(A : X \to Y),\ \text{closed}(A) \wedge \text{linear}(A) \Rightarrow \text{bounded}(A)$	ClosedGraph	ZF + DC
Thm 6	GaussGreen : $\int_U u_{x_i} = \int_{\partial U} u\nu^i \wedge \int_U \text{div u} = \int_{\partial U} \text{u} \cdot \nu$	Gauss − Green	ZF + DC
Thm 7	IntParts : $\forall(u : \bar{U} \to \mathbb{R}),\ \forall(v : \bar{U} \to \mathbb{R}),\ \int_U u_{x_i} v = -\int_U u v_{x_i} + \int_{\partial U} uv\nu^i$	IntegrationbyParts (nD)	ZF + DC
Thm 8	GreenI : $\forall(u : \bar{U} \to \mathbb{R}),\ \int_U \Delta u = \int_{\partial U} \partial u/\partial \nu$	GreenFirstIdentity	ZF + DC
Thm 9	GreenII : $\forall(u : \bar{U} \to \mathbb{R}),\ \forall(v : \bar{U} \to \mathbb{R}),\ \int_U \nabla v \cdot \nabla u = -\int_U u\Delta v + \int_{\partial U} u\partial v/\partial \nu$	GreenSecondIdentity	ZF + DC
Thm 10	GreenIII : $\forall(u, v : \bar{U} \to \mathbb{R}),\ \int_U (u\Delta v - v\Delta u) = \int_{\partial U} (u\partial v/\partial \nu - v\partial u/\partial \nu)$	GreenThirdIdentity	ZF + DC
Thm 11	Coarea : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(u : \mathbb{R}^n \to \mathbb{R}),\ \int f\lvert\nabla u\rvert = \int (\int_{\{u=r\}} f\, dS)\, dr$	CoareaFormula	ZF + DC
Thm 12	Polar : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(x_0 : \mathbb{R}^n),\ \int f = \int_0^\infty (\int_{\partial B(x_0,r)} f\, dS)\, dr$	PolarCoordinates	ZF + DC
Thm 13	MovingRegions : $\frac{d}{d\tau} \int_{U(\tau)} f = \int_{\partial U(\tau)} f\text{v} \cdot \nu + \int_{U(\tau)} f_\tau$	ReynoldsTransport	ZF + DC
Thm 14	MollifierProps : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ f^\epsilon \in C^\infty \wedge f^\epsilon \to f$ a.e. $\wedge$ unif on compacta if $C(f) \wedge L^p_{\text{loc}}$ conv.	MollifierProperties	ZF + DC
Thm 15	LebesgueExistence : $\exists(\mathcal{M} : \text{SigmaAlgebra}),\ \exists(\lvert\cdot\rvert : \mathcal{M} \to \mathbb{R}_{\geq 0} \cup \{\infty\}),\ \mathcal{M}\ \sigma-\text{alg.} \wedge \lvert B\rvert = \text{vol}(B) \wedge \sigma-\text{additive}$	LebesgueMeasure Existence	ZF + DC
Thm 16	MonotoneConv : $\forall(f : \mathbb{N} \to (\mathbb{R} \to \mathbb{R})),\ 0 \leq f_k \uparrow\ \Rightarrow \int \lim f_k = \lim \int f_k$	MonotoneConvergence	ZF + DC
Thm 17	Fatou : $\int \liminf f_k \leq \liminf \int f_k$	Fatou	ZF + DC
Thm 18	DominatedConv : $\forall(f : \mathbb{N} \to (\mathbb{R} \to \mathbb{R})),\ \forall(g : \mathbb{R} \to \mathbb{R}),\ \lvert f_k\rvert \leq g \wedge L^1(g) \wedge f_k \to f$ a.e. $\Rightarrow \int f_k \to \int f$	Dominated Convergence	ZF + DC
Thm 19	Holder : $\forall(u : \mathbb{R}^n \to \mathbb{R}),\ \forall(v : \mathbb{R}^n \to \mathbb{R}),\ \int \lvert uv\rvert \leq \lVert u\rVert_{L^p} \lVert v\rVert_{L^q} \wedge 1/p + 1/q = 1$	Holder	ZF + DC
Thm 20	Minkowski : $\forall(u : \mathbb{R}^n \to \mathbb{R}),\ \forall(v : \mathbb{R}^n \to \mathbb{R}),\ \lVert u + v\rVert_{L^p} \leq \lVert u\rVert_{L^p} + \lVert v\rVert_{L^p}$	Minkowski	ZF + DC
Thm 21	GenHolder : $\int \lvert u_1 \cdots u_m\rvert \leq \prod \lVert u_k\rVert_{L^{p_k}}$	GeneralizedHolder	ZF + DC
Thm 22	Interpolation : $\forall(u : \mathbb{R}^n \to \mathbb{R}),\ \lVert u\rVert_{L^r} \leq \lVert u\rVert^\theta_{L^s} \lVert u\rVert^{1-\theta}_{L^t}$	Interpolation	ZF + DC
Thm 23	LebesgueDiff : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \forall(x_0 : \mathbb{R}^n),\ \text{avgB}(f, x_0, r) \to f(x_0)$ a.e.	Lebesgue Differentiation	ZF + DC
Thm 24	Egoroff : a.e.-conv. $\Rightarrow$ unif. off ϵ	Egoroff	ZF + DC
Thm 25	Bochner : $\forall(\text{f} : X),\ \text{summable}(\text{f}) \Leftrightarrow \text{summable}(\lVert\text{f}\rVert) \wedge \lVert \int \text{f}\rVert \leq \int \lVert\text{f}\rVert$	Bochner	ZF + DC
Thm 26	Pettis : $\forall(\text{f} : X),\ \text{strong meas.}(\text{f}) \Leftrightarrow \text{weak meas.}(\text{f}) \wedge \text{almost sep. valued}(\text{f})$	Pettis	ZF + DC
Thm 27	Fredholm : $\forall(K : H \to H),\ \text{CompactOp}(K) \Rightarrow \dim N(I-K) < \infty \wedge \text{ClosedOp}(R(I-K)) \wedge R(I-K) = \text{Sperp}(N(I-K^*)) \wedge (N(I-K) = \{0\} \Leftrightarrow R(I-K) = H) \wedge \dim N(I-K) = \dim N(I-K^*)$	FredholmAlternative	ZF + DC
Thm 28	SpectrumCompact : $\forall(K : H \to H),\ \dim H = \infty \wedge \text{CompactOp}(K) \Rightarrow 0 \in \sigma(K) \wedge \sigma(K) \setminus \{0\} = \sigma_p(K) \setminus \{0\} \wedge \text{finite-or-tends0}(\sigma(K) \setminus \{0\})$	SpectrumofCompact Op.	ZF + DC
Thm 29	SupportingHyperplanes : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \text{convex}(f) \Rightarrow \forall(x : \mathbb{R}^n),\ \exists(r : \mathbb{R}^n),\ \forall(y : \mathbb{R}^n),\ f(y) \geq f(x) + r \cdot (y - x)$	SupportingHyperplanes	ZF
Thm 30	Jensen : $\forall(f : \mathbb{R}^n \to \mathbb{R}),\ \text{convex}(f) \Rightarrow f(\text{avgE}(\text{u}, U)) \leq \text{avgE}(f \circ \text{u}, U)$	Jensen	ZF

Thm 31	**ArzelaAscoli** : $\forall (f : \mathbb{N} \to (\mathbb{R} \to \mathbb{R})),$ unif. bdd$(f) \wedge$ equicont$(f) \Rightarrow$ unif. conv. subseq.	Arzela – Ascoli	ZF + DC
Thm 32	**WeakCompact** : reflexive$(X) \wedge$ bdd$(u) \Rightarrow$ weakly conv. subseq.	WeakCompactness	ZFC
Thm 33	**RellichKondrachov** : $\forall (U$ bdd$),\ W^{1,p}(U) \hookrightarrow\hookrightarrow L^q(U)$ compactly for $q < p^*$	Rellich – Kondrachov	ZF + DC
Thm 34	**CompactAdjoint** : $\forall (K : H \to H),\ K$ compact $\Rightarrow K^*$ compact	CompactnessofAdjoint	ZFC
Thm 35	**SpectralThm** : $\forall (H :$ Hilbert$),\ \forall (S : H \to H),$ Separable$(H) \wedge$ CompactOp$(S) \wedge$ Symmetric$(S) \Rightarrow \exists (w : \mathbb{N} \to H),$ ONB-of-eigenvectors(S, w)	SpectralTheorem	ZF + DC
Thm 36	**PoissonExistence** : $\forall (f \in H^{-1}(U)),\ \exists (u \in H_0^1(U)),\ -\Delta u = f$ weak	DirichletExistence	ZF + DC
Thm 37	**PoissonUniqueness** : $\forall (u_1, u_2 \in H_0^1(U)),$ WeakSol$(u_1) \wedge$ WeakSol$(u_2) \Rightarrow u_1 = u_2$ a.e.	DirichletUniqueness	ZF + DC
Thm 38	**DirichletPrinciple** : $\forall (u \in H_0^1(U)),\ u$ weak sol $\Leftrightarrow u$ minimizes $I[w]$ over $H_0^1(U)$	DirichletPrinciple	ZF + DC
Thm 39	**EllipticInteriorReg** : $\forall (u \in H^1,\ f \in L^2,\ -\Delta u = f),\ \forall (V \subset\subset U),\ u \in H^2_{\mathrm{loc}}(U) \wedge \|u\|_{H^2(V)} \le C(\|f\|_{L^2} + \|u\|_{H^1})$	InteriorRegularity	ZF + DC
Thm 40	**EllipticBoundaryReg** : $\forall (\partial U \in C^2,\ f \in L^2,\ g \in H^{3/2}(\partial U)),\ \exists! (u \in H^2(U)),\ -\Delta u = f \wedge$ trace$(u) = g \wedge \|u\|_{H^2} \le C(\|f\|_{L^2} + \|g\|_{H^{3/2}})$	BoundaryRegularity	ZF + DC
Thm 41	**MaximumPrinciple** : $\forall (u \in C^2(U) \cap C^0(\bar{U}),\ \Delta u \ge 0),\ \max_{\bar{U}} u = \max_{\partial U} u$	WeakMaximum Principle	ZF
Thm 42	**HopfLemma** : $\forall (u \in C^2(U) \cap C^0(\bar{U}),\ \Delta u \ge 0,\ x_0 \in \partial U,\ u(x_0) > u(x)\ \forall x \in U),\ \partial u/\partial \nu(x_0) > 0$	HopfLemma	ZF
Thm 43	**StrongMaxPrinciple** : $\forall (u \in C^2(U),\ \Delta u \ge 0),$ (u attains interior max) $\Rightarrow u$ constant	StrongMaximum Principle	ZF
Thm 44	**HeatWeakExistence** : $\forall (f \in L^2(0,T;H^{-1}),\ u_0 \in L^2),\ \exists! (u$ weak sol heat$),\ u(0) = u_0 \wedge \|u\|^2_{L^2(0,T;H_0^1)} + \|u_t\|^2_{L^2(0,T;H^{-1})} \le C_T(\|f\|^2 + \|u_0\|^2)$	HeatWeakExistence	ZF + DC
Thm 45	**HeatEnergyDecay** : $\forall (u$ weak sol heat$,\ f = 0),\ \|u(t)\|_{L^2} \le e^{-\lambda_1 t}\|u_0\|_{L^2}$	HeatEnergyDecay	ZF + DC
Thm 46	**WaveEnergyConservation** : $\forall (u$ weak sol wave$,\ f = 0),\ \frac{d}{dt}(\frac{1}{2}\|u_t\|^2 + \frac{1}{2}\|\nabla u\|^2) = 0$	WaveEnergy Conservation	ZF + DC
Thm 47	**WaveWeakExistence** : $\forall (f \in L^2(0,T;L^2),\ u_0 \in H_0^1,\ u_1 \in L^2),\ \exists! (u$ weak sol wave$),\ u(0) = u_0 \wedge u_t(0) = u_1$	WaveWeakExistence	ZF + DC
Thm 48	**DuhamelPrinciple** : $\forall (f \in L^2(0,T;H^{-1})),\ u(t) = \int_0^t S(t-s)f(s)\,ds$ weak sol heat with $u_0 = 0$	DuhamelPrinciple	ZF + DC

6. Proofs

Proof of Lemma 1 (FTC)

1	$C^1([a,b])(f)$	hypothesis	Hyp
2	$F(x) := \int_a^x f'(t)\,dt$	def	def
3	$F'(x) = f'(x)$	FTC step	Eq(2)
4	$(F - f)' = 0$	derivative	Eq(3)

5	$\exists (C : \mathbb{R}),\ \forall (x : [a,b]),\ (F - f)(x) = C$	MVT	Pos:1(4)
6	$F(a) = 0 \Rightarrow C = -f(a)$	initial value	Eq(2, 5)
7	$F(b) = f(b) - f(a)$	endpoint	Eq(5, 6)
8	$\top$	close	$\rightarrow$ I(1, . . . , 7)

Proof of Lemma 2 (Taylor)

1	$C^2(f) \wedge (x : \mathbb{R}) \wedge (z : \mathbb{R})$	hypothesis	Hyp
2	$\int_z^x (x-t)f'' = [-(x-t)f']_z^x + \int_z^x f'$	IBP	Pos:3(1)
3	$\int_z^x (x-t)f'' = \int_z^x f' - (x-z)f'(z)$	simplify	Eq(2)
4	$\int_z^x f' = f(x) - f(z)$	FTC	Lem:1(3)
5	$f(x) = f(z) + (x-z)f'(z) + \int_z^x (x-t)f''$	rearrange	Eq(3, 4)
6	$\exists (\xi : [\min(x,z), \max(x,z)]),\ \int_z^x (x-t)f'' = \frac{1}{2}f''(\xi)(x-z)^2$	weighted mean	Pos:2(1)
7	$f(x) = f(z) + (x-z)f'(z) + \frac{1}{2}f''(\xi)(x-z)^2$	conclusion	Eq(5, 6)
8	$\top$	close	$\rightarrow$ I(1, . . . , 7)

Proof of Lemma 3 (ProductRule)

1	$C^1(u) \wedge C^1(v)$	hypothesis	Hyp
2	$(uv)(x+h) - (uv)(x) = u(x+h)v(x+h) - u(x)v(x)$	expand	def
3	$= [u(x+h) - u(x)]v(x+h) + u(x)[v(x+h) - v(x)]$	add zero	Eq(2)
4	$\frac{(uv)(x+h)-(uv)(x)}{h} = \frac{u(x+h)-u(x)}{h}v(x+h) + u(x)\frac{v(x+h)-v(x)}{h}$	divide	Eq(3)
5	$\lim_{h\to 0} = u'(x)v(x) + u(x)v'(x)$	limit	Eq(1, 4)
6	$\top$	close	$\rightarrow$ I(1, . . . , 5)

Proof of Lemma 4 (ChainRule)

1	$C^1(f) \wedge C^1(g)$	hypothesis	Hyp
2	$F(y) := \begin{cases} \frac{f(y) - f(g(x))}{y - g(x)}, & \text{if } y \neq g(x) \\ f'(g(x)), & \text{if } y = g(x) \end{cases}$	aux	def
3	$C(F)$ at $g(x)$	continuity	Eq(1, 2)
4	$(f \circ g)(x+h) - (f \circ g)(x) = F(g(x+h)) \cdot (g(x+h) - g(x))$	exact	Eq(2)
5	$\lim_{h\to 0} F(g(x+h)) = f'(g(x))$	comp. cont.	Eq(3)
6	$\lim_{h\to 0}(g(x+h) - g(x))/h = g'(x)$	hypothesis	Hyp

7	$\lim_{h\to 0} \frac{(f\circ g)(x+h)-(f\circ g)(x)}{h} = f'(g(x))g'(x)$	limit	Eq(4, 5, 6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Lemma 5 (JacobiFormula)

1	$C^1(A) \wedge \det A(t) \neq 0$	hypothesis	Hyp
2	$\det(A + \epsilon B) = \det A + \epsilon \mathsf{tr}(\mathsf{adj}(A)B) + O(\epsilon^2)$	expansion	def
3	$\mathsf{adj}(A) = \det A \cdot A^{-1}$	adjugate	def
4	$\frac{\det(A+\epsilon B)-\det A}{\epsilon} = \det A \mathsf{tr}(A^{-1}B) + O(\epsilon)$	quotient	Eq(2, 3)
5	$\frac{d}{dt} \det A = \det A \mathsf{tr}(A^{-1}A')$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 6 (JacobiFormulaAdj)

1	$C^1(A)$	hypothesis	Hyp
2	$\det(A + \epsilon B) = \det A + \epsilon \mathsf{tr}(\mathsf{adj}(A)B) + O(\epsilon^2)$	expansion	def
3	$\frac{d}{dt} \det A = \mathsf{tr}(\mathsf{adj}(A)A')$	conclusion	Eq(2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Lemma 7 (Cauchy)

1	$(a : \mathbb{R}) \wedge (b : \mathbb{R})$	hypothesis	Hyp
2	$0 \leq (a - b)^2$	square nonneg	Ax:5(1)
3	$0 \leq a^2 - 2ab + b^2$	expand	Eq(2)
4	$2ab \leq a^2 + b^2$	add	addMon(3)
5	$ab \leq \frac{1}{2}a^2 + \frac{1}{2}b^2$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 8 (CauchyEps)

1	$a, b \in \mathbb{R},\ \epsilon > 0$	hypothesis	Hyp
2	$ab = ((2\epsilon)^{1/2}a)(b/(2\epsilon)^{1/2})$	factor	Eq
3	$ab \leq \frac{1}{2}((2\epsilon)^{1/2}a)^2 + \frac{1}{2}(b/(2\epsilon)^{1/2})^2$	Cauchy	Lem:7(2)
4	$ab \leq \epsilon a^2 + \frac{b^2}{4\epsilon}$	conclusion	Eq(3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Lemma 9 (SecondDerivCriterion)

1	$C^2(f) \wedge f'' \geq 0$	hypothesis	Hyp
2	$x, y \in \mathbb{R},\ t \in [0,1],\ z := (1-t)x + ty$	intro	intro
3	$f(x) = f(z) + f'(z)(x-z) + \frac{1}{2}f''(\xi_1)(x-z)^2$	Taylor	Lem:2(1, 2)
4	$f(x) \geq f(z) + f'(z)(x-z)$	lower	Eq(3, 1)
5	$f(y) \geq f(z) + f'(z)(y-z)$	Taylor + lower	Lem:2(1, 2)
6	$(1-t)f(x) + tf(y) \geq f(z)$	combine	addMon(4, 5)
7	$f(\tau x + (1-\tau)y) \leq \tau f(x) + (1-\tau)f(y)$	conclusion	Def:64(2, 6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Lemma 10 (expconvex)

1	$f(t) := e^t \wedge f'' = e^t > 0$	def	def
2	$C^2(f) \wedge f'' \geq 0$	regularity	Eq(1)
3	convex(f)	convexity	Lem:9(2)
4	$e^{\tau x + (1-\tau)y} \leq \tau e^x + (1-\tau)e^y$	conclusion	Def:64(3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Lemma 11 (Young)

1	$1 < p, q < \infty,\ 1/p + 1/q = 1,\ a, b > 0$	hypothesis	Hyp
2	$t := 1/p,\ x := \log a^p,\ y := \log b^q$	intro	intro
3	$e^{tx + (1-t)y} \leq te^x + (1-t)e^y$	expconvex	Lem:10(2)
4	$e^{(1/p)\log a^p + (1/q)\log b^q} \leq \frac{1}{p}e^{\log a^p} + \frac{1}{q}e^{\log b^q}$	substitute	Eq(3)
5	$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 12 (YoungEps)

1	$a, b > 0,\ \epsilon > 0,\ 1 < p, q < \infty,\ 1/p + 1/q = 1$	hypothesis	Hyp
2	$ab = ((\epsilon p)^{1/p} a)(b/(\epsilon p)^{1/p})$	factor	Eq
3	$ab \leq \epsilon a^p + \frac{b^q}{q(\epsilon p)^{q/p}}$	Young	Lem:11(2)
4	$C(\epsilon) := (\epsilon p)^{-q/p}/q$	define	def
5	$ab \leq \epsilon a^p + C(\epsilon) b^q$	conclusion	Eq(3, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 13 (CauchySchwarz)

1	$x, y \in \mathbb{R}^n$	hypothesis	Hyp
2	$y = 0 \Rightarrow \lvert x \cdot y\rvert = 0 = \lvert x\rvert\lvert y\rvert$	trivial	Eq(1)
3	$y \neq 0 \Rightarrow 0 \leq \lvert x - \epsilon y\rvert^2 = \lvert x\rvert^2 - 2\epsilon x \cdot y + \epsilon^2 \lvert y\rvert^2$	nonneg	Ax:5(1)
4	$2\epsilon x \cdot y \leq \lvert x\rvert^2 + \epsilon^2 \lvert y\rvert^2$	add	addMon(3)
5	$\epsilon = \lvert x\rvert / \lvert y\rvert \Rightarrow x \cdot y \leq \lvert x\rvert\lvert y\rvert$	choose	Eq(4)
6	$-x \cdot y \leq \lvert x\rvert\lvert y\rvert$	substitute	Eq(3, . . . , 5)
7	$\lvert x \cdot y\rvert \leq \lvert x\rvert\lvert y\rvert$	conclusion	Eq(2, 5, 6)
8	$\top$	close	→ I(1, . . . , 7)

Proof of Lemma 14 (CauchySchwarzA)

1	$A > 0 \wedge A^T = A \wedge x, y \in \mathbb{R}^n$	hypothesis	Hyp
2	$(u, v) \mapsto u^T A v$ inner product	inner	Def:47(1)
3	$\lvert x^T A y\rvert \leq (x^T A x)^{1/2} (y^T A y)^{1/2}$	Cauchy − Schwarz	Lem:13(2)
4	$\top$	close	→ I(1, . . . , 3)

Proof of Lemma 15 (BanachFP)

1	contraction$(T) \wedge$ complete(X)	hypothesis	Hyp
2	$\exists \theta < 1,\ d(Tx, Ty) \leq \theta d(x, y)$	def	def(1)
3	$x_{n+1} := T(x_n)$	iterate	def
4	$d(x_{n+1}, x_n) \leq \theta^n d(x_1, x_0)$	geometric	Eq(2, 3)
5	$d(x_n, x_m) \leq \frac{\theta^n}{1-\theta} d(x_1, x_0)$	bound	Eq(4)
6	Cauchy(x_n)	Cauchy	Def:54(5)
7	$\exists x^*,\ x_n \to x^*$	limit	Ax:4(6)
8	$Tx^* = x^* \wedge \forall y,\ Ty = y \to y = x^*$	fixed + unique	Eq(2, 3, 7)
9	$\top$	close	→ I(1, . . . , 8)

Proof of Lemma 16 (SeparatingHyperplane)

1	convex$(C) \wedge$ closed$(C) \wedge p \in \partial C$	hypothesis	Hyp
2	$\exists (p_k),\ p_k \notin C \wedge p_k \to p$	exterior	def(1)
3	$\exists (\pi_k \subset C),\ \lVert p_k - \pi_k\rVert =$ dist(p_k, C)	projection	∃E(2)
4	$\forall (z \in C),\ (p_k - \pi_k) \cdot (z - \pi_k) \leq 0$	convexity	Def:64(1, 3)

5	$u_k := \frac{p_k - \pi_k}{\|p_k - \pi_k\|},\ \|u_k\| = 1$	normalize	def
6	$\exists u_{k_j} \to u,\ \|u\| = 1$	BW	Pos:4(5)
7	$\forall (z \in C),\ u \cdot (z - p) \le 0$	conclusion	Eq(4, 5, 6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Lemma 17 (Baire)

1	complete(X) $\wedge$ U_n open dense	hypothesis	Hyp
2	$\bar{B}(x_0, r_0) \subset U_1$	nested	def(1)
3	$\bar{B}(x_n, r_n) \subset U_n \cap B(x_{n-1}, r_{n-1}),\ r_n \to 0$	recursion	$\exists$E(2)
4	Cauchy(x_n) $\Rightarrow \exists x^*$	complete	Ax:4(3)
5	$x^* \in \bigcap_n U_n$	conclusion	Eq(3, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 18 (OpenMap)

1	$T \in \mathcal{L}(X, Y)$ $\wedge$ surjective(T)	hypothesis	Hyp
2	$X = \bigcup_n \bar{B}(0, n),\ Y = \bigcup_n T(\bar{B}(0, n))$	cover	Eq(1)
3	$\exists n,\ \overline{T(B(0, n))}$ nonempty interior	Baire	Lem:17(2)
4	$\exists r > 0,\ \bar{B}_Y(0, r) \subseteq T(\bar{B}(0, n))$	ball	$\exists$E(3)
5	$T(B(0, 1)) \supseteq B_Y(0, r/n) \Rightarrow$ open(T)	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 19 (HardyLittlewood)

1	$f \in L^1_{\text{loc}}$	hypothesis	Hyp
2	$M(f)(x) := \sup_{r>0}$ avgB($\|f\|, x, r$)	maximal	Def:30(1)
3	$\|\text{WeakL}^1(M(f))\| \le C\|f\|_{L^1}$	weak type	def(2)
4	avgB($\|f - f(x)\|, x, r$) $\to 0$ a.e.	conclusion	Eq(1, 3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Lemma 20 (Riesz)

1	$u^* \in H^*$	hypothesis	Hyp
2	$N(u^*)$ closed subspace	kernel	def
3	$u^* = 0 \Rightarrow u = 0$	trivial	Eq(1)
4	$u^* \neq 0 \Rightarrow \exists w \in N(u^*)^\perp,\ \|w\| = 1$	perp	Def:37(2)

5	$u := \overline{u^*(w)}\, w \Rightarrow \forall v,\ u^*(v) = (u, v)$	representation	Def:35(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 21 (LaxMilgram)

1	coercive$(a) \wedge$ bounded$(a) \wedge (f : H^*)$	hypothesis	Hyp
2	$\exists!(A : H \to H),\ \forall(u : H),\ \forall(v : H),\ a(u, v) = (Au, v)$	Riesz	Lem:20(2)
3	$\|Au\| \geq c\|u\|$	coercivity	Eq(1, 2)
4	A injective $\wedge$ bdd(A)	injectivity	Eq(3)
5	$R(A)$ closed $\wedge$ A surjective	closed range	Eq(1, 4)
6	$\exists!(u : H),\ Au = h \wedge h =$ Riesz(f)	existence	$\exists$E(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 22 (BoundsSpectrum)

1	S bounded self $-$ adjoint	hypothesis	Hyp
2	$\eta > M \Rightarrow (\eta I - S)$ coercive $\Rightarrow$ bijective	coercive	Lem:21(1)
3	$\eta < m \Rightarrow (S - \eta I)$ coercive $\Rightarrow$ bijective	coercive	Lem:21(1)
4	$\sigma(S) \subset [m, M]$	bounded	$\forall$E(2, 3)
5	$M \in \sigma(S) \wedge m \in \sigma(S)$	attained	RAA + Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 23 (WeakDerivRules)

1	$u, w \in W^{1,p}(U),\ \phi \in \mathcal{D}(U)$	hypothesis	Hyp
2	$\int_U (u + w)D\phi = \int_U uD\phi + \int_U wD\phi$	linearity	Eq(1)
3	$= -(\int_U Du \cdot \phi + \int_U Dw \cdot \phi) = -(\int_U (Du + Dw)\phi)$	WeakDeriv def	Def:75(1)
4	WeakDeriv$(D(u + w)) = Du + Dw$	conclusion	Eq(3)
5	WeakDeriv$(D(u\phi)) = u'\phi + uD\phi$	product	Lem:3(1)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 24 (Poincare)

1	$\partial U \in C^0,\ u \in H_0^1(U)$	hypothesis	Hyp
2	u extends by 0 across ∂U	def	Def:78(1)
3	$u(x) = \int_{-\infty}^{x_1} u_{x_1}(t, x')\, dt$	FTC	Lem:1(2)
4	$\|u(x)\|^2 \leq \operatorname{diam}(U) \int_{-\infty}^{x_1} \|u_{x_1}\|^2\, dt$	Cauchy $-$ Schwarz	Lem:13(3)

5	$\int_U \lvert u\rvert^2 \le C_P(U)^2 \int_U \lvert\nabla u\rvert^2$	integrate	Eq(4)
6	$\lVert u\rVert_{L^2} \le C_P(U)\lVert\nabla u\rVert_{L^2}$	conclusion	Eq(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 25 (TraceInequality)

1	$u \in H^1(U)$	hypothesis	Hyp
2	$\exists \mathsf{Tr} : H^1 \to L^2(\partial U)$ lin. cont., $\mathsf{Tr} \circ \mathsf{Ext} = \mathrm{id}$	trace	Pos:7
3	$\lVert\mathsf{Tr}(u)\rVert_{L^2(\partial U)} \le \lVert\mathsf{Tr}\rVert \cdot \lVert u\rVert_{H^1(U)}$	bounded	Eq(2)
4	$\lVert u\rVert_{L^2(\partial U)} = \lVert\mathsf{Tr}(u)\rVert_{L^2(\partial U)}$	identification	def(2)
5	$\lVert u\rVert_{L^2(\partial U)} \le C_T\lVert u\rVert_{H^1(U)}$	conclusion	Eq(3, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Lemma 26 (SobolevEmbedding)

1	$u \in W^{1,p}(U),\ p > n$	hypothesis	Hyp
2	$\lVert u\rVert_{C^{0,1-n/p}(U)} \le C_M\lVert u\rVert_{W^{1,p}(U)}$	Morrey	Pos:6
3	$u \in C^{0,1-n/p}(U)$	embedding	Eq(2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Lemma 27 (EnergyEstimateElliptic)

1	$f \in L^2(U),\ u$ weak sol	hypothesis	Hyp
2	$B[u, u] = \langle f, u\rangle$	def	Def:81(1)
3	$\lVert\nabla u\rVert_{L^2}^2 = \langle f, u\rangle$	Dirichlet form	Def:79(2)
4	$\langle f, u\rangle \le \lVert f\rVert_{L^2}\lVert u\rVert_{L^2} \le C_P(U)\lVert f\rVert_{L^2}\lVert\nabla u\rVert_{L^2}$	Holder + Poincare	Lem:24(3)
5	$\lVert\nabla u\rVert_{L^2} \le C_P(U)\lVert f\rVert_{L^2}$	cancel	Eq(4)
6	$\lVert u\rVert_{H_0^1} \le C\lVert f\rVert_{L^2}$	conclusion	Lem:24(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Lemma 28 (EllipticInteriorReg-Step)

1	$u \in H^1,\ -\Delta u = f \in L^2$	hypothesis	Hyp
2	$V \subset\subset U,\ \zeta \in \mathcal{D}(U),\ \zeta \equiv 1$ on V	cutoff	def
3	$B[\zeta u, \zeta u] = -\int_U f\zeta^2 u - \int_U u\nabla\zeta \cdot \nabla(\zeta u)$	weak sol	Def:81(1)
4	$\lVert D^2(\zeta u)\rVert_{L^2} \le C(\lVert f\rVert_{L^2} + \lVert u\rVert_{H^1})$	energy	Lem:27(3)

5	$\top$	close	$\to$ I$(1,\dots,4)$

Proof of Theorem 1 (BanachAlaoglu)

1	Banach(X)	hypothesis	Hyp
2	$\bar{B}_{X^*}(0,1) \subset \prod_{x\in X}[-\|x\|,\|x\|]$	embed	def
3	$\prod_x[-\|x\|,\|x\|]$ compact	Tychonoff	Pos:5
4	$\bar{B}_{X^*}(0,1)$ closed	closed	def(2)
5	$\bar{B}_{X^*}(0,1)$ weak*-compact	conclusion	Eq$(3,4)$
6	$\top$	close	$\to$ I$(1,\dots,5)$

Proof of Theorem 2 (Kakutani)

1	reflexive(X)	hypothesis	Hyp
2	$\bar{B}_X(0,1) \subseteq X = (X^*)^*$	reflexivity	Def:65(1)
3	$\bar{B}_{X^{**}}(0,1)$ weak*-compact	Banach — Alaoglu	Thm:1
4	$\bar{B}_X(0,1)$ weakly compact	conclusion	Eq$(2,3)$
5	$\top$	close	$\to$ I$(1,\dots,4)$

Proof of Theorem 3 (InverseFT)

1	$Jf(x_0) \neq 0$, WLOG $x_0 = 0$, $f(0) = 0$, $Df(0) = I$	normalize	def
2	$f(x) = x + R(x)$, $\|DR\| \leq \frac{1}{2}$	decompose	def(1)
3	$\|f(x) - f(y)\| \geq \frac{1}{2}\|x - y\|$	estimate	Pos:1(2)
4	$\forall z$ near 0, $\exists x$, $f(x) = z$	surjectivity	Lem:15(2)
5	$g := f^{-1}$, $Dg(z) = Df(g(z))^{-1}$	chain rule	Lem:4(4)
6	$f \in C^k \Rightarrow g \in C^k$	conclusion	$\forall$I(5)
7	$\top$	close	$\to$ I$(1,\dots,6)$

Proof of Theorem 4 (ImplicitFT)

1	$f(x_0,y_0) = z_0$, $D_y f(x_0,y_0)$ invertible	hypothesis	Hyp
2	$F(x,y) := (f(x,y),x)$	define	def
3	$\det DF = \pm \det D_y f \neq 0$	Jacobian	def(2)
4	F locally invertible, $F^{-1} \in C^1$	inverse FT	Thm:3(3)
5	$g(x) :=$ 2nd$(F^{-1}(z_0,x)) \Rightarrow f(x,g(x)) = z_0$	solve	Eq(4)
6	$f \in C^k \Rightarrow g \in C^k$	conclusion	$\forall$E(4)

7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 5 (ClosedGraph)

1	closed(A) $\wedge$ linear(A) $\wedge$ Banach(X, Y)	hypothesis	Hyp
2	$G(A) := \{(u, Au)\}$ closed in $X \times Y$	graph	def
3	$X \times Y$ Banach	product	Def:62(1)
4	$P : G(A) \to X,\ (u, Au) \mapsto u$ bijective	projection	def(2)
5	P^{-1} bounded	open map	Lem:18(4)
6	Bdd(A)	conclusion	Def:39(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 6 (GaussGreen)

1	U smooth, $u \in C^1(\bar{U})$	hypothesis	Hyp
2	$\{\zeta_k\}$ partition of unity, $y = \Phi(x)$ flattens	partition	Def:49(1)
3	$\int_{y_n>0} \partial_{y_n}(\zeta u) = \int_{y_n=0} \zeta u$	FTC	Lem:1(2)
4	$\int_U (\zeta u)_{x_n} = \int_{\partial U} \zeta u \nu^n$	transform	Eq(2, 3)
5	$\int_U u_{x_i} = \int_{\partial U} u\nu^i \wedge \int_U \operatorname{div} \mathrm{u} = \int_{\partial U} \mathrm{u} \cdot \nu$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 7 (IntParts)

1	$u, v \in C^1(\bar{U})$	hypothesis	Hyp
2	$\int_U (uv)_{x_i} = \int_{\partial U} uv\nu^i$	Gauss — Green	Thm:6(1)
3	$(uv)_{x_i} = u_{x_i} v + uv_{x_i}$	product	Lem:3(1)
4	$\int_U u_{x_i} v = -\int_U uv_{x_i} + \int_{\partial U} uv\nu^i$	conclusion	Eq(2, 3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Theorem 8 (GreenI)

1	$u \in C^2(\bar{U})$	hypothesis	Hyp
2	$\int_U u_{x_i x_i} = \int_{\partial U} u_{x_i} \nu^i$	IBP	Thm:7(1)
3	$\int_U \Delta u = \int_{\partial U} \partial u / \partial \nu$	conclusion	Def:48(2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Theorem 9 (GreenII)

1	$u, v \in C^2(\bar{U})$	hypothesis	Hyp
2	$\int_U u_{x_i} v_{x_i} = -\int_U u v_{x_i x_i} + \int_{\partial U} u v_{x_i} \nu^i$	IBP	Thm:7(1)
3	$\int_U \nabla v \cdot \nabla u = -\int_U u \Delta v + \int_{\partial U} u \partial v / \partial \nu$	conclusion	Def:20(2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Theorem 10 (GreenIII)

1	$u, v \in C^2(\bar{U})$	hypothesis	Hyp
2	$\int_U \nabla u \cdot \nabla v = -\int_U v \Delta u + \int_{\partial U} v \partial u / \partial \nu$	Green II	Thm:9(1)
3	$\int_U (u \Delta v - v \Delta u) = \int_{\partial U} (u \partial v / \partial \nu - v \partial u / \partial \nu)$	conclusion	Eq(2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Theorem 11 (Coarea)

1	Lipschitz$(u) \wedge$ Measurable(f)	hypothesis	Hyp
2	$\exists f_k$ simple, $f_k \to f$ a.e.	approx	Def:33(1)
3	$\int f_k \lvert\nabla u\rvert = \int (\int_{\{u=r\}} f_k \, dS) \, dr$	simple case	def(2)
4	$\int f_k \lvert\nabla u\rvert \uparrow \quad \int f \lvert\nabla u\rvert \wedge \int (\int f_k) \uparrow \quad \int (\int f)$	monotone	Thm:16(3)
5	$\int f \lvert\nabla u\rvert = \int (\int_{\{u=r\}} f \, dS) \, dr$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 12 (Polar)

1	$x_0 = 0$, $u(x) := \lvert x\rvert$, $\lvert\nabla u\rvert = 1$ a.e.	normalize	def
2	$\int f \lvert\nabla u\rvert = \int_0^\infty (\int_{\partial B(0,r)} f \, dS) \, dr$	Coarea	Thm:11(1)
3	$\int f = \int_0^\infty (\int_{\partial B(0,r)} f \, dS) \, dr$	conclusion	Eq(1, 2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Theorem 13 (MovingRegions)

1	$X(\tau, \cdot) : U(0) \to U(\tau)$, $\partial_\tau X = \mathrm{v}$	flow	def
2	$\int_{U(\tau)} f = \int_{U(0)} f(X(\tau, y), \tau) J \, dy$, $J := \det \partial_y X$	change var.	def
3	$\partial_\tau J = J \mathrm{div}(\mathrm{v})$	Jacobi	Lem:5(2)
4	$\partial_\tau [fJ] = [f_\tau + \nabla f \cdot \mathrm{v}] J + f J \mathrm{div}(\mathrm{v})$	product	Lem:3(3)

5	$\frac{d}{d\tau}\int_{U(\tau)} f = \int_{U(\tau)}[f_\tau + \operatorname{div}(f\mathbf{v})]$	transport	Eq(2, 4)
6	$= \int_{\partial U(\tau)} f\mathbf{v}\cdot\nu + \int_{U(\tau)} f_\tau$	conclusion	Thm:6(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 14 (MollifierProps)

1	$f \in L^1_{\mathrm{loc}},\ f^\epsilon = \eta_\epsilon * f$	hypothesis	Def:51(1)
2	$D^\alpha f^\epsilon(x) = \int D^\alpha \eta_\epsilon(x-y) f(y)\,dy$	derivative	def(1)
3	$f^\epsilon \in C^\infty$	smoothness	$\forall$I(2)
4	$\lvert f^\epsilon(x) - f(x)\rvert \le C\,\mathrm{avgB}(\lvert f - f(x)\rvert) \to 0$	Hardy – Littlewood	Lem:19(1)
5	$f^\epsilon \to f$ a.e., $f \in C \Rightarrow$ unif on compacta	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 15 (LebesgueExistence)

1	Box(I), vol(I), outmeas	def	def
2	outmeas monotone $\wedge$ countably subadditive	properties	Eq(1)
3	$\mathcal{M}$ Carathodory sigma – algebra	def	def
4	SigmaAlgebra($\mathcal{M}$)	sigma	Def:58(3)
5	$\lvert\cdot\rvert :=$ outmeas$\lvert_{\mathcal{M}}$ countably additive, $\lvert B\rvert = \mathrm{vol}(B)$	measure	Eq(2, 4)
6	open(U), closed(F) $\in \mathcal{M}$	conclusion	Eq(3)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 16 (MonotoneConv)

1	$0 \le f_1 \le f_2 \le \cdots,\ f := \sup_k f_k$	hypothesis	Hyp
2	$\int f_k \uparrow\ \ L \le \int f$	limit	def(1)
3	$0 \le s \le f$ simple, $E_k := \{f_k \ge (1-\epsilon)s\} \uparrow\ \ X$	approx	Def:33(1)
4	$(1-\epsilon)\int_{E_k} s \le L \Rightarrow \int s \le L$	monotone	Eq(2, 3)
5	$\int f = L = \lim \int f_k$	conclusion	Eq(2, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 17 (Fatou)

1	$f_k \ge 0$ measurable	hypothesis	Hyp
2	$g_k := \inf_{j\ge k} f_j$ nondecreasing	infimum	def

3	$\int g_k \uparrow \ \int \liminf f_k$	monotone conv	Thm:16(2)
4	$\int g_k \le \inf_{j \ge k} \int f_j \Rightarrow \int \liminf f_k \le \liminf \int f_k$	conclusion	Eq(2, 3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Theorem 18 (DominatedConv)

1	$\lvert f_k \rvert \le g$, $g \in L^1$, $f_k \to f$ a.e.	hypothesis	Hyp
2	$\int (g + f) \le \liminf \int (g + f_k) \Rightarrow \int f \le \liminf \int f_k$	Fatou	Thm:17(1)
3	$\int (g - f) \le \liminf \int (g - f_k) \Rightarrow \limsup \int f_k \le \int f$	Fatou	Thm:17(1)
4	$\int f \le \liminf \int f_k \le \limsup \int f_k \le \int f$	squeeze	Eq(2, 3)
5	$\int f_k \to \int f$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 19 (Holder)

1	$\|u\|_{L^p} = \|v\|_{L^q} = 1$	hypothesis	Hyp
2	$\lvert uv \rvert \le \lvert u \rvert^p / p + \lvert v \rvert^q / q$ a.e.	Young	Lem:11(1)
3	$\int \lvert uv \rvert \le \|u\|_{L^p}^p / p + \|v\|_{L^q}^q / q = 1$	integrate	Eq(2)
4	$\int \lvert uv \rvert \le \|u\|_{L^p} \|v\|_{L^q}$	homogeneity	Eq(1, 3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Theorem 20 (Minkowski)

1	$u, v \in L^p$	hypothesis	Hyp
2	$\|u + v\|_{L^p}^p \le \int \lvert u + v \rvert^{p-1} (\lvert u \rvert + \lvert v \rvert)$	triangle	Eq
3	$\int \lvert u + v \rvert^{p-1} \lvert u \rvert \le \|u + v\|_{L^p}^{p-1} \|u\|_{L^p}$	Holder	Thm:19(2)
4	$\int \lvert u + v \rvert^{p-1} \lvert v \rvert \le \|u + v\|_{L^p}^{p-1} \|v\|_{L^p}$	Holder	Thm:19(2)
5	$\|u + v\|_{L^p} \le \|u\|_{L^p} + \|v\|_{L^p}$	conclusion	Eq(2, 3, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 21 (GenHolder)

1	$1 \le p_1, \ldots, p_m \le \infty$, $\sum 1/p_k = 1$	hypothesis	Hyp
2	$m = 2 \Rightarrow$ GenHolder	base	Thm:19(1)
3	IH : $m - 1$ case, $q_j := p_j / (1 - 1/p_1)$	induction	Hyp
4	$\int \lvert u_1 \cdots u_m \rvert \le \|u_1\|_{L^{p_1}} \|u_2 \cdots u_m\|_{L^{1/(1-1/p_1)}}$	Holder	Thm:19(3)

5	$\|u_2 \cdots u_m\|_{L^{1/(1-1/p_1)}} \le \prod_{k\ge 2} \|u_k\|_{L^{p_k}}$	**IH**	**Eq(3)**
6	$\int \|u_1 \cdots u_m\| \le \prod \|u_k\|_{L^{p_k}}$	**conclusion**	**Eq(4, 5)**
7	$\top$	**close**	**$\to$ I(1, . . . , 6)**

Proof of Theorem 22 (Interpolation)

1	$1/r = \theta/s + (1-\theta)/t$	**hypothesis**	**Hyp**
2	$\int \|u\|^r = \int \|u\|^{\theta r} \|u\|^{(1-\theta)r} \le (\int \|u\|^s)^{\theta r/s} (\int \|u\|^t)^{(1-\theta)r/t}$	**Holder**	**Thm:19(1)**
3	$\|u\|_{L^r} \le \|u\|_{L^s}^{\theta} \|u\|_{L^t}^{1-\theta}$	**conclusion**	**Eq(1, 2)**
4	$\top$	**close**	**$\to$ I(1, . . . , 3)**

Proof of Theorem 23 (LebesgueDiff)

1	$f \in L^1_{\mathrm{loc}},\ u(x) := \mathsf{avgB}(\|f - f(x)\|, x, r)$	**hypothesis**	**Def:30**
2	$\mathsf{avgB}(\|f - f(x_0)\|, x_0, r) \to 0$ **a.e.**	**Hardy – Littlewood**	**Lem:19(1)**
3	$\mathsf{avgB}(f, x_0, r) \to f(x_0)$ **a.e.**	**conclusion**	**Eq(2)**
4	$\top$	**close**	**$\to$ I(1, . . . , 3)**

Proof of Theorem 24 (Egoroff)

1	$f_k \to f$ **a.e.,** $\|A\| < \infty$	**hypothesis**	**Hyp**
2	$A_j^m := \bigcap_{k\ge j} \{\|f_k - f\| < 1/m\}$	**sets**	**def**
3	$A_j^m \uparrow \ \tilde{A}^m$ **a.e.** A	**increasing**	**Eq(1, 2)**
4	$\exists j(m),\ \|A \setminus A_{j(m)}^m\| < \epsilon/2^m$	**measure**	**$\exists$E(3)**
5	$E := \bigcap_m A_{j(m)}^m \Rightarrow \|A \setminus E\| \le \epsilon \wedge$ **unif on** E	**conclusion**	**Eq(4)**
6	$\top$	**close**	**$\to$ I(1, . . . , 5)**

Proof of Theorem 25 (Bochner)

1	**strongly measurable(f)**	**hypothesis**	**Hyp**
2	$\exists \mathrm{s}_k$ **simple,** $\mathrm{s}_k \to \mathrm{f}$	**approx**	**Def:53(1)**
3	$\int \|\mathrm{s}_k - \mathrm{f}\| \to 0$	**dominated**	**Thm:18(2)**
4	**summable(f)** $\Leftrightarrow$ **summable(**$\|\mathrm{f}\|$**)** $\wedge \|\int \mathrm{f}\| \le \int \|\mathrm{f}\|$	**conclusion**	**Eq(3)**
5	$\top$	**close**	**$\to$ I(1, . . . , 4)**

Proof of Theorem 26 (Pettis)

1	strongly meas(f) $\Rightarrow$ weakly meas + almost sep	forward	Def:53(1)
2	weakly meas + almost sep $\Rightarrow$ strongly meas(f)	reverse	$\wedge$I(1)
3	strongly meas $\Leftrightarrow$ weakly meas + almost sep	conclusion	Eq(1, 2)
4	$\top$	close	$\to$ I(1, . . . , 3)

Proof of Theorem 27 (Fredholm)

1	CompactOp(K)	hypothesis	Hyp
2	$\dim N(I-K) = \infty \Rightarrow \exists$ONB, $Ku_k = u_k$, $\|u_k - u_l\| = \sqrt{2}$	contradiction	Def:36(1)
3	$\|Ku_k - Ku_l\| = \sqrt{2} \Rightarrow \neg$compact($K$)	noncompact	$\perp$I(2)
4	$\dim N(I-K) < \infty$, $R(I-K)$ closed, $R(I-K) =$ Sperp($N(I-K^*)$)	combine	Eq(2, 3)
5	$N(I-K) = \{0\} \Leftrightarrow R(I-K) = H$, $\dim N(I-K) = \dim N(I-K^*)$	conclusion	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 28 (SpectrumCompact)

1	CompactOp(K) $\wedge \dim H = \infty$	hypothesis	Hyp
2	$0 \notin \sigma(K) \Rightarrow I = K \circ K^{-1}$ compact $\Rightarrow \neg \dim H = \infty$	contradiction	Def:56(1)
3	$0 \in \sigma(K)$	zero	RAA(2)
4	$\eta \in \sigma(K) \setminus \{0\} \Rightarrow$ eigenvalue	Fredholm	Thm:27(4)
5	$\sigma(K) \setminus \{0\}$ tends to 0	conclusion	Eq(3, 4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 29 (SupportingHyperplanes)

1	convex(f) $\wedge\, x \in \mathbb{R}^n$	hypothesis	Hyp
2	epi(f) convex, $(x, f(x)) \in \partial$epi(f)	epigraph	Def:64(1)
3	$\exists (r, -\alpha) \neq 0$, $(r, -\alpha) \cdot ((z,t) - (x, f(x))) \leq 0$	separating	Lem:16(1, 2)
4	$\alpha > 0 \Rightarrow f(z) \geq f(x) + r \cdot (z - x)$	conclusion	Eq(3)
5	$\forall x\ \exists r,\ \forall y,\ f(y) \geq f(x) + r \cdot (y - x)$	quantify	$\forall$I(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 30 (Jensen)

1	convex$(f)\wedge$ open bounded$(U)\wedge$ summable(u)	hypothesis	Hyp
2	$\forall p\,\exists r\,\forall q,\ f(q)\ge f(p)+r\cdot(q-p)$	support	Thm:29(1)
3	$p:=$ avgE$(\mathrm{u},U),\ q:=\mathrm{u}(x)$	intro	def(2)
4	$\int_U f(\mathrm{u})\ge \lvert U\rvert f(p)+r\cdot(\int \mathrm{u}-\lvert U\rvert p)=\lvert U\rvert f(p)$	integrate	Def:32(3)
5	$f($avgE$(\mathrm{u},U))\le$ avgE$(f\circ\mathrm{u},U)$	conclusion	Eq(4)
6	$\top$	close	$\to$ I$(1,\dots,5)$

Proof of Theorem 31 (ArzelaAscoli)

1	unif. bdd$(f_k)\wedge$ equicont(f_k)	hypothesis	Hyp
2	$\exists\{x_1,\dots,x_N\}$ epsilon $-$ net in K	net	$\exists$E(1)
3	$\exists\{k_j\},\ \forall i,\ f_{k_j}(x_i)$ converges	diagonal	$\exists$E(2)
4	$\lvert f_{k_j}(x)-f_{k_l}(x)\rvert\le 2\epsilon+\lvert f_{k_j}(x_i)-f_{k_l}(x_i)\rvert$	equicont	Eq$(2,3)$
5	$f_{k_j}\to f$ unif	conclusion	Eq(4)
6	$\top$	close	$\to$ I$(1,\dots,5)$

Proof of Theorem 32 (WeakCompact)

1	reflexive$(X)\wedge$ bdd(u_k)	hypothesis	Hyp
2	$\bar B(0,M)$ weakly compact	Kakutani	Thm:2(1)
3	$\exists\{u_{k_j}\},\ u_{k_j}\rightharpoonup u$	subsequence	Eq(2)
4	$\top$	close	$\to$ I$(1,\dots,3)$

Proof of Theorem 33 (RellichKondrachov)

1	U bounded, $q<p^*$	hypothesis	Hyp
2	$\{u_k\}$ bdd in $W^{1,p}$	hypothesis	Hyp
3	$\exists u_{k_j}\rightharpoonup u$ in $W^{1,p}$	weak compact	Thm:32(1)
4	$\lVert u_{k_j}\rVert_{W^{1,p}}\le C$	uniform bound	Eq$(2,3)$
5	$\lVert u_{k_j}-u\rVert_{L^q}\to 0$	diagonal + embedding	Eq$(3,4)$
6	$\top$	close	$\to$ I$(1,\dots,5)$

Proof of Theorem 34 (CompactAdjoint)

1	CompactOp(K)	hypothesis	Hyp
2	$\|u_k\|$ bdd $\Rightarrow \exists u_{k_j} \rightharpoonup u$	weak compact	Thm:32(1)
3	$\|K^*(u_{k_j}-u)\|^2 = (KK^*(u_{k_j}-u), u_{k_j}-u)$	adjoint	Def:67(1)
4	$KK^*(u_{k_j}-u) \to 0 \Rightarrow \|K^*(u_{k_j}-u)\| \to 0$	compactness	Def:72(1)
5	CompactOp(K^*)	conclusion	Def:72(4)
6	$\top$	close	$\to$ I$(1,\dots,5)$

Proof of Theorem 35 (SpectralThm)

1	Separable$(H) \wedge$ CompactOp$(S) \wedge$ Symmetric(S)	hypothesis	Hyp
2	$H_k := N(S-\eta_k I)$ finite dim, $H_k \perp H_l$	eigenspaces	Thm:27(1)
3	$\tilde{H} := \overline{\bigoplus H_k}$, $\tilde{S} := S\|_{\tilde{H}^\perp}$, $\sigma(\tilde{S}) = \{0\}$	decompose	Def:37(2)
4	$(\tilde{S}u, u) = 0\ \forall u \Rightarrow \tilde{S} = 0$	trivial	Lem:22(4)
5	$\tilde{H}^\perp \subset N(S) \subset \tilde{H} \Rightarrow \tilde{H} = H$	completeness	Eq$(3,4)$
6	S has an ONB of eigenvectors	conclusion	Eq$(2,5)$
7	$\top$	close	$\to$ I$(1,\dots,6)$

Proof of Theorem 36 (PoissonExistence)

1	$f \in H^{-1}(U)$	hypothesis	Hyp
2	$B[u,v] := \int_U \nabla u \cdot \nabla v$	Dirichlet form	Def:79(1)
3	B bounded on $H_0^1 \times H_0^1$, $\|B\| \le 1$	Cauchy $-$ Schwarz	Lem:13(1)
4	B coercive on H_0^1, $B[u,u] \ge (1+C_P(U)^2)^{-1}\|u\|_{H^1}^2$	Poincare	Lem:24(1)
5	$f \in H^{-1} = (H_0^1)^*$	def	Def:57(1)
6	$\exists!(u : H_0^1),\ \forall(v : H_0^1),\ B[u,v] = \langle f, v\rangle$	Lax $-$ Milgram	Lem:21(3,4,5)
7	$-\Delta u = f$ weak	conclusion	Def:81(6)
8	$\top$	close	$\to$ I$(1,\dots,7)$

Proof of Theorem 37 (PoissonUniqueness)

1	$u_1, u_2 \in H_0^1$ weak sol	hypothesis	Hyp
2	$w := u_1 - u_2 \in H_0^1$, $B[w,v] = 0\ \forall v$	linearity	Def:79(1)
3	$B[w,w] = 0$	choose $v = w$	Eq(2)
4	$B[w,w] \ge \theta\|w\|_{H^1}^2$	coercivity	Def:81(1)

5	$\|w\|_{H^1} = 0$	combine	Eq(3, 4)
6	$u_1 = u_2$ a.e.	conclusion	Eq(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 38 (DirichletPrinciple)

1	$u \in H_0^1(U)$, weak sol	hypothesis	Hyp
2	$w \in H_0^1$ arbitrary	hypothesis	Hyp
3	$I[u + w] - I[u] = \frac{1}{2}\|\nabla w\|^2 + B[u, w] - \langle f, w\rangle$	expand	Def:84(1)
4	$B[u, w] = \langle f, w\rangle$	weak sol	Def:81(1)
5	$I[u + w] - I[u] = \frac{1}{2}\|\nabla w\|^2 \geq 0$	simplify	Eq(3, 4)
6	u minimises I	conclusion	Eq(5)
7	converse : $I[u] \leq I[u + w]\ \forall w \Rightarrow B[u, w] = \langle f, w\rangle$	first variation	Eq(3)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Theorem 39 (EllipticInteriorReg)

1	$u \in H^1,\ f \in L^2,\ -\Delta u = f$	hypothesis	Hyp
2	$V \subset\subset U,\ \zeta \in \mathcal{D}(U),\ \zeta \equiv 1$ on V	cutoff	def
3	$B[\zeta u, \zeta u] = -\int_U f\zeta^2 u - \int_U u\nabla\zeta \cdot \nabla(\zeta u)$	weak sol	Def:81(1)
4	$\|\nabla(\zeta u)\|_{L^2}^2 \leq C(\|f\|_{L^2}^2 + \|u\|_{L^2}^2)$	energy	Lem:27(3)
5	$w := \zeta u,\ B[w, w] \geq \theta\|\nabla w\|^2$	coercivity	Def:81(1)
6	$B[D_i^h w, D_i^h w] = -\int_U D_i^h f \cdot D_i^h w - \int_U D_i^h(u\nabla\zeta) \cdot D_i^h w$	difference quotient	Eq(3)
7	$\|D_i^h w\|_{L^2} \leq C(\|f\|_{L^2} + \|u\|_{H^1})$	uniform in h	Eq(5, 6)
8	$\|D^2(\zeta u)\|_{L^2} \leq C(\|f\|_{L^2} + \|u\|_{H^1})$	combine	Eq(7)
9	$u \in H^2_{\mathrm{loc}}(U)$	localise	Eq(8)
10	$\top$	close	$\to$ I(1, . . . , 9)

Proof of Theorem 40 (EllipticBoundaryReg)

1	$\partial U \in C^2,\ f \in L^2,\ g \in H^{3/2}(\partial U)$	hypothesis	Hyp
2	$\exists!(u \in H^2),\ -\Delta u = f \wedge \mathrm{Tr}(u) = g \wedge \|u\|_{H^2} \leq C(\|f\|_{L^2} + \|g\|_{H^{3/2}})$	boundary regularity	Pos:8
3	$\top$	close	$\to$ I(1, 2)

Proof of Theorem 41 (MaximumPrinciple)

1	$u \in C^2(U) \cap C^0(\bar{U}),\ \Delta u \geq 0$	hypothesis	Hyp
2	$\epsilon > 0,\ u_\epsilon := u + \epsilon \lvert x \rvert^2$	perturb	def
3	$\Delta u_\epsilon \geq 2n\epsilon > 0$	compute	Def:48(2)
4	u_ϵ cannot attain interior max	RAA + 2nd deriv	Eq(3)
5	$\max_{\bar{U}} u_\epsilon = \max_{\partial U} u_\epsilon$	boundary	Eq(4)
6	$\epsilon \to 0 \Rightarrow \max_{\bar{U}} u \leq \max_{\partial U} u$	limit	Eq(5)
7	$\top$	close	$\to$ I(1, . . . , 6)

Proof of Theorem 42 (HopfLemma)

1	$u \in C^2(U) \cap C^0(\bar{U}),\ \Delta u \geq 0,\ u(x_0) > u(x)\ \forall x \in U$	hypothesis	Hyp
2	$v(x) := e^{-\lambda \lvert x \rvert^2} - e^{-\lambda R^2},\ \Delta v > 0$	aux	def
3	$w_\epsilon := u - u(x_0) + \epsilon v$	def	def
4	$\Delta w_\epsilon > 0,\ w_\epsilon < 0$ on $\partial B(x_0, r) \setminus \{x_0\}$	max principle	Thm:41(1)
5	$\partial u / \partial \nu(x_0) \geq -\epsilon \partial v / \partial \nu(x_0) > 0$	limit	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 43 (StrongMaxPrinciple)

1	$u \in C^2(U),\ \Delta u \geq 0,\ u$ attains interior max at x_0	hypothesis	Hyp
2	u not constant $\Rightarrow \exists B(x_0, r) \subset U,\ \sup_{\partial B} u < u(x_0)$	contradiction	def(1)
3	$\partial u / \partial \nu(x_0) > 0$	Hopf	Thm:42(1, 2)
4	contradiction with interior max $\Rightarrow u$ constant	conclusion	RAA(1, 3)
5	$\top$	close	$\to$ I(1, . . . , 4)

Proof of Theorem 44 (HeatWeakExistence)

1	$f \in L^2(0, T; H^{-1}),\ u_0 \in L^2$	hypothesis	Hyp
2	$\{\psi_k\}$ ONB of $H_0^1,\ u_m := \sum_{k \leq m} d_k(t) \psi_k$	Galerkin	Def:36(1)
3	$(u_m', \psi_k) + B[u_m, \psi_k] = \langle f, \psi_k \rangle,\ u_m(0) = P_m u_0$	ODE system	def(2)
4	$\frac{d}{dt} \lVert u_m \rVert^2 + 2B[u_m, u_m] = 2\langle f, u_m \rangle$	energy identity	Eq(3)
5	$\lVert u_m(t) \rVert^2 + \int_0^T \lVert \nabla u_m \rVert^2 \leq C(\lVert f \rVert^2 + \lVert u_0 \rVert^2)$	integrate + Gronwall	Eq(4)
6	$\lVert u_m' \rVert_{L^2(0,T;H^{-1})} \leq C(\lVert f \rVert + \lVert u_0 \rVert)$	dual norm	Eq(3, 5)

7	$\exists u_{m_j} \rightharpoonup u$ in $L^2(0,T;H_0^1)$, $u'_{m_j} \rightharpoonup u_t$ in $L^2(0,T;H^{-1})$	compactness	Thm:32(4)
8	u weak sol, $u(0)=u_0$	pass to limit	Eq(3, 7)
9	uniqueness via energy on $w=u_1-u_2$	Gronwall	Eq(4)
10	$\top$	close	$\to$ I(1, . . . , 9)

Proof of Theorem 45 (HeatEnergyDecay)

1	u weak sol heat, $f=0$	hypothesis	Hyp
2	$\frac{d}{dt}\|u\|_{L^2}^2 = -2\|\nabla u\|_{L^2}^2$	energy	Def:82(1)
3	$\|\nabla u\|_{L^2}^2 \geq \lambda_1 \|u\|_{L^2}^2$	Poincare + eigenvalue	Lem:24(3)
4	$\frac{d}{dt}\|u\|^2 \leq -2\lambda_1\|u\|^2$	combine	Eq(2, 3)
5	$\|u(t)\|_{L^2} \leq e^{-\lambda_1 t}\|u_0\|_{L^2}$	Gronwall	Eq(4)
6	$\top$	close	$\to$ I(1, . . . , 5)

Proof of Theorem 46 (WaveEnergyConservation)

1	u weak sol wave, $f=0$	hypothesis	Hyp
2	$E(t) := \frac{1}{2}\|u_t\|^2 + \frac{1}{2}\|\nabla u\|^2$	energy	def
3	$u_t \in L^2(0,T;L^2)$, $u_{tt} \in L^2(0,T;H^{-1})$	regularity	Def:83(1)
4	$E'(t) = (u_{tt}, u_t) + (\nabla u, \nabla u_t)$	chain rule	Eq(2)
5	$(u_{tt}, u_t) + B[u, u_t] = \langle 0, u_t\rangle = 0$	wave eq	Def:83(1)
6	$E'(t) = 0$ for a.e. t	combine	Eq(4, 5)
7	$E(t) = E(0)$ for a.e. t	integrate	Eq(6)
8	$\top$	close	$\to$ I(1, . . . , 7)

Proof of Theorem 47 (WaveWeakExistence)

1	$f \in L^2(0,T;L^2)$, $u_0 \in H_0^1$, $u_1 \in L^2$	hypothesis	Hyp
2	$\{\psi_k\}$ ONB of H_0^1	eigenbasis	Def:36(1)
3	$u_m := \sum_{k\leq m} d_k(t)\psi_k$, $u_m(0) = P_m u_0$, $u'_m(0) = P_m u_1$	Galerkin	def(2)
4	$(u''_m, \psi_k) + B[u_m, \psi_k] = \langle f, \psi_k\rangle$	ODE system	def(3)
5	$E_m(t) := \frac{1}{2}\|u'_m\|^2 + \frac{1}{2}\|\nabla u_m\|^2$, $E'_m(t) = \langle f, u'_m\rangle$	energy identity	Thm:46(1)
6	$E_m(t) \leq E_m(0) + \int_0^T \|f\|\|u'_m\|$	Gronwall	Eq(5)
7	$E_m(t) \leq C(\|f\|_{L^2}^2 + \|u_0\|_{H^1}^2 + \|u_1\|^2)$	uniform in m	Eq(6)
8	$\exists u_{m_j} \rightharpoonup^* u$, $u'_{m_j} \rightharpoonup^* u_t$, $u''_{m_j} \rightharpoonup^* u_{tt}$	compactness	Thm:32(4)

9	u solves $u_{tt} - \Delta u = f$ weak, $u(0) = u_0$, $u_t(0) = u_1$	pass to limit	Eq(4, 8)
10	uniqueness via energy on $w = u_1 - u_2$	Gronwall	Eq(7)
11	$\top$	close	$\rightarrow$ I(1, . . . , 10)

Proof of Theorem 48 (DuhamelPrinciple)

1	$f \in L^2(0, T; H^{-1})$	hypothesis	Hyp
2	$S(t)$ C_0-semigroup with $S'(t) = \Delta S(t)$	generator	Pos:9
3	$u(t) := \int_0^t S(t-s) f(s)\, ds$	def	def
4	$u_t(t) = S(t-t)f(t) + \int_0^t S'(t-s)f(s)\, ds = f(t) + \int_0^t \Delta S(t-s) f(s)\, ds$	differentiate	Eq(2, 3)
5	$= f(t) + \Delta u(t)$	linearity	Eq(3, 4)
6	$u_t - \Delta u = f,\ u(0) = 0$	conclusion	Eq(3, 5)
7	$\top$	close	$\rightarrow$ I(1, . . . , 6)

7. Scripts

Each tree records, for one result, the entries used by its proof, transitively closed. Names coincide with the names that appear in the statement tables. Leaves are either postulates (unproved statements admitted inside proofs), axioms of the ordered field $\mathbb{R}$ (primitive structure), or primitive definitions. In the trees below the prefix `Ld` carries the depth of the node (root at `L0`) and the name is indented `2*d` spaces after the pipe, so every chain can be read straight down a column of increasing depth labels. Trees are typeset in a two-column grid, each tree inside its own `minipage` so that the depth-ordered chains can be compared side by side.

Foundational leaves (postulates).

```
MVT   MVTI   IBP   BW   Tych
Morrey   Trace
EllipticBoundaryReg-Step
HeatSemigroupGenerator
```

Foundational leaves (axioms of the ordered field R)

```
ax:R-comm       ax:R-assoc
ax:R-order      ax:R-complete
ax:R-mul        ax:R-add
```

Elementary definition leaves used in the trees.

```
Def:littleo         Def:InnerProduct
Def:Convex          Def:SigmaAlgebra
Def:Simple          Def:Sperp
Def:perp            Def:avg-ball
Def:avg-set         Def:moll-f
Def:Bdd             Def:Banach
Def:Reflexive       Def:adjoint
Def:CompactOp       Def:ONB
Def:sigma           Def:Lap
Def:grad            Def:Ckbnd
Def:Symmetric       Def:StronglyMeasurable
Def:WeakDeriv       Def:SobolevWkp
Def:SobolevHk       Def:H01
Def:EllipticForm    Def:Elliptic
Def:PoissonWeakSol  Def:ParabolicWeakSol
Def:WaveWeakSol     Def:DirichletEnergy
```

Analytic core.

```
L0 | FTC
L1 |   MVT
```

```
L0 | Taylor
L1 |   IBP
L1 |   MVTI
L1 |   FTC
L2 |     MVT
```

Hilbert space layer.

```
L0 | Riesz
L1 |   Def:Sperp
L1 |   Def:perp
```

```
L0 | LaxMilgram
L1 |   Riesz
L2 |     Def:Sperp
L2 |     Def:perp
```

PDE existence and regularity.

```
L0 | Poincare
L1 |   FTC
L1 |   CauchySchwarz
L2 |     ax:R-mul
```

```
L0 | PoissonUniqueness
L1 |   Def:EllipticForm
L1 |   Def:PoissonWeakSol
```

```
L0 | PoissonExistence
L1 |   Def:EllipticForm
L1 |   Def:PoissonWeakSol
L1 |   Poincare
L2 |     FTC
L2 |     CauchySchwarz
L1 |   LaxMilgram
L2 |     Riesz
L3 |       Def:Sperp
L3 |       Def:perp
```

```
L0 | DirichletPrinciple
L1 |   Def:DirichletEnergy
L1 |   PoissonExistence
L2 |     LaxMilgram
L3 |       Riesz
L4 |         Def:Sperp
L4 |         Def:perp
```

```
L0 | EllipticInteriorReg
L1 |   EnergyEstimateElliptic
L2 |     Poincare
L3 |       FTC
L3 |       CauchySchwarz
L1 |   Def:Lap
```

```
L0 | EllipticBoundaryReg
L1 |   EllipticBoundaryReg-Step
L1 |   Trace
```

Compactness and maximum principles.

```
L0 | RellichKondrachov
L1 |   WeakCompact
L2 |     Kakutani
L3 |       BanachAlaoglu
L4 |         Tych
```

```
L0 | MaximumPrinciple
L1 |   Def:Lap
```

```
L0 | HopfLemma
L1 |   MaximumPrinciple
L2 |     Def:Lap
```

```
L0 | StrongMaxPrinciple
L1 |   HopfLemma
L2 |     MaximumPrinciple
L3 |       Def:Lap
```

Heat and wave equations.

```
L0 | HeatWeakExistence
L1 |   Poincare
L2 |     FTC
L2 |     CauchySchwarz
L1 |   WeakCompact
L2 |     Kakutani
L3 |       BanachAlaoglu
L4 |         Tych
```

```
L0 | HeatEnergyDecay
L1 |   ParabolicWeakSol
L1 |   Poincare
L2 |     FTC
L2 |     CauchySchwarz
```

```
L0 | WaveEnergyConservation
L1 |   Def:WaveWeakSol
L1 |   Def:EllipticForm
```

```
L0 | WaveWeakExistence
L1 |   WaveEnergyConservation
L1 |   WeakCompact
L2 |     Kakutani
L3 |       BanachAlaoglu
L4 |         Tych
```

```
L0 | DuhamelPrinciple
L1 |   HeatSemigroupGenerator
L1 |   HeatWeakExistence
L2 |     Poincare
L2 |     WeakCompact
L3 |       Kakutani
L4 |         BanachAlaoglu
L5 |           Tych
```

8. Conclusion

The document presents the dependency-ordered exposition of PDE theory (prerequisites plus the solution layer). All entries are organised into six parts (Axioms, Postulates, Definitions, Statements, Proofs, Scripts) and the choice strength of every result is recorded in the class column as ZF, ZF+DC, or ZFC. Within the Statements part, Lemmas and Theorems form a single dependency layer, so long as each entry appears only after all entries it depends on. In particular Riesz is stated among the Lemmas because LaxMilgram depends on it, LaxMilgram precedes BoundsSpectrum for the same reason, WaveEnergyConservation precedes WaveWeakExistence, and RellichKondrachov is stated among the Theorems (after WeakCompact) because its proof uses the weak compactness principle.

Two closure properties are worth flagging. First, the boundary regularity of the Dirichlet problem is admitted as a postulate (EllipticBoundaryReg-Step) rather than derived from the interior regularity theorem, because the latter only yields H^2_{loc}. Second, the heat semigroup $S(t)$ used in the proof of Duhamel's principle is introduced through the postulate HeatSemigroupGenerator, which records the generator identity $S'(t) = \Delta S(t)$ as an external input rather than deriving it from the Galerkin construction. Both postulates are tagged ZF+DC and are the only assumptions of their respective proofs that are not established elsewhere in the document.

Every lemma and theorem stated in the Statements section has a corresponding proof table in the Proofs section, and the proofs are presented in exactly the same order as the statements.

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